06 Ch 06 – Portfolio Management

Asked in last 3 papers: Sep 2025 single-index risk split + two-stock portfolio (6m); Jan 2026 Sharpe optimal cut-off (8m) + CAPM/SML MCQs (6m); May 2026 two-factor APT long/short (6m), Sharpe optimal with given C* + CAPM (6m), SGB MCQs (6m).

1. Introduction

→ Investor never puts all funds in one security; a **well diversified portfolio** optimises risk-return; a portfolio can **reduce risk without diluting returns**.

→ **Risk** = possibility of actual return deviating from expected/average return; measured by **standard deviation / variance** of return.

1.1 Activities in Portfolio Management

→ (a) **Selection** of securities; (b) construction of all **feasible portfolios** from them; (c) deciding **weights** so that it is the **optimal portfolio** for that investor's risk appetite.

1.2 Objectives of Portfolio Management

→ **Security/safety of principal** (intact sum and its purchasing power); **stability of income** (plan reinvestment/consumption); **capital growth** (reinvest in growth securities).

→ **Marketability** (ease of buying/selling, flexibility); **liquidity** (nearness to money, to seize opportunities); **diversification** (reduce risk of loss of capital/income across securities and industries); **favourable tax status** (minimise tax to raise effective yield).

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TYK T2: objectives of portfolio management → answer is para 1.2 (seven objectives above).

2. Phases of Portfolio Management (2.1–2.5)

Phase	What it does (SM points)
2.1 Security analysis	Initial phase: risk-return of individual securities and their **correlation**; find under-priced (buy) and over-priced (sell). **Fundamental analysis**: EPS, payout, competition, market share, management, industry factors → intrinsic value vs CMP. **Technical analysis**: price movements systematic, past patterns predict future; ignores fundamentals. Select across sectors with no strong positive correlation so unsystematic risk is eliminated. **Random Walk**: price moves random, neither analysis gives sustained gains; **EMH**: no sustained long-term gains; markets never perfectly efficient.
2.2 Portfolio analysis	Combine securities; each portfolio's risk-return is **not the aggregate** of constituents — computed from their risk-return and **pair-wise correlations**. All portfolios from the chosen set = **Feasible Portfolios**.
2.3 Portfolio selection	Identify **Efficient Portfolios** (highest return for same/lower risk, or lowest risk for same/higher return), then the **Optimal Portfolio** suiting risk appetite (Markowitz).
2.4 Portfolio revision	Monitor so it does not lose optimality (economy, new securities); add/delete securities; also investor changes (extra funds, risk appetite, need of cash). Must be scientific and objective — as important as analysis and selection.
2.5 Portfolio evaluation	Measure actual return realised and risk borne over the period; identifies weaknesses in the process. Process is ongoing: analysis → construction → revision → evaluation.

3. Portfolio Theories — 3.1 Traditional Approach

→ **Investor's study:** age, health, responsibilities, other assets, portfolio needs; need for income, capital maintenance, liquidity; attitude to risk; taxation status.

→ **Objectives:** maximise wealth subject to risk; higher risk → higher expected return.

→ **Investment strategy:** balance fixed interest vs equities; high payout vs high growth companies; income of growth portfolio; income tax vs capital gains tax; transaction cost vs gains from rapid switching; retain liquidity for bargains.

→ **Diversification** reduces volatility; balance equities and fixed interest.

→ **Selection:** intrinsic value vs market (fundamental) or past prices (technical); expert advice; inside information; newspaper tipsters; companies with good asset backing, dividend growth, earnings record, quality management.

→ Followed by most Indian brokers. Says "don't put all money in one basket" — Markowitz showed only **particular kinds** of securities give diversification benefit; don't pick blindly.

3.2 Modern Approach (Markowitz Model or Risk-Return Optimization)

→ Harry Markowitz (early 1950s), **father of Modern Portfolio Theory**; **Mean Variance Analysis** (1952).

→ Two planks: (1) adding securities with **low or negative correlation** reduces risk; (2) higher return needs higher risk. Risk of an individual asset hardly matters — what matters is its **contribution to overall portfolio risk**.

4. Risk Analysis — 4.1 Types of Risk

→ Risk = variation of actual (realised) return from expected return; stable returns (Govt securities) = low risk; equity = risky.

→ **Total risk = Systematic risk + Unsystematic risk.** Systematic: external, macro, affects many securities, uncontrollable. Unsystematic: internal, micro, company-specific, largely controllable.

4.1.1 Systematic (non-diversifiable)	4.1.2 Unsystematic ("idiosyncratic", diversifiable)
Interest rate risk: prices move inversely with rates; long-term securities vary more than short-term.	**Business risk:** variability of operating profit; measured by **operating leverage**.
Purchasing power (inflation) risk: nominal return = real return + inflation premium; unexpected inflation erodes return; more for bonds/fixed income, less for equity (dividends rise, capital gains).	**Financial (leveraged) risk:** due to debt; expressed by **debt-equity ratio**; highly geared → risk of winding up.
Market risk: share moves with the market — bullish (general rise) / bearish (general fall); seen in BSE Sensex, NSE index.	

4.2 Diversion of Risk

→ Combining many securities **diversifies away unsystematic risk**; total risk falls as number of securities rises; beyond a limit the portfolio contains **only systematic risk**.

4.3 Risk & Return

→ Return of a share = dividend + capital gain over the period. Risk cannot be measured without reference to return.

Formula	Terms / logic	SM example figures
R = (Forecasted dividend + Forecasted end of the period stock price) ÷ Initial investment − 1; same as Returns = $[D_1 + (P_1 - P_0)] \div P_0 \times 100$ (SM Illus 6)	D_1 dividend; P_1 end price; P_0 initial price (investment). Holding-period return.	X Ltd: P_0 ₹ 12.00, D ₹ 0.50, P_1 ₹ 17.50 → (0.50 + 17.50)/12 − 1 = **0.5 = 50%**.

4.3.1 Expected Return

Formula	Terms / logic	SM example figures
(a) With probabilities: $\bar{X} = \Sigma X_i \cdot p(X_i)$	X_i possible return; $p(X_i)$ its probability. Probability-weighted average.	Returns 0.20/0.30/0.40/0.50/0.60, p 0.2/0.2/0.4/0.1/0.1 → **0.37 = 37%**.
(b) Historical: $\bar{x} = \Sigma x_i \div n$	Simple arithmetic mean of past returns.	X Ltd 5 years 9.50, 7.00, 11.00, −6.00, 2.50 → Σ 24.00 ÷ 5 = **4.80%**.

Note (app-written): the SM lists the possible returns as 20, 30, 50, 60, 70% (p. 13–14), but the worked table uses 0.20, 0.30, 0.40, 0.50, 0.60 with the same probabilities to get 0.37. ICAI's 37% follows the worked table.

4.3.2 Risk

Formula	Terms / logic	SM example figures
(a) $\sigma^2 = \Sigma (X_i - \bar{X})^2 \cdot p(X_i)$	Deviation from mean, squared, weighted by probability. $\sigma = \sqrt{\sigma^2}$ (positive root).	Same data: variance **0.0141**, $\sigma = \sqrt{0.0141}$ = **11.87%**.
(b) $\sigma^2 = \Sigma (x_i - \bar{x})^2 \div n$	Historical data: divide by number of observations instead of multiplying by probabilities.	Σ dev^2 187.30 ÷ 5 = **37.46**; σ = **6.12%**.

→ Range of return = mean ± σ (e.g. 11% ± 3% → 8% to 14%). Other measures: range, semi-variance, mean absolute deviation — SD most accepted.

→ SD = **total risk**. Unsystematic risk can be eliminated by combining with negatively correlated securities, so the **relevant risk is systematic** (not diversifiable).

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TYK P1: cost ₹ 120, no dividend; prices 115–140 → returns −4.17, 0, 4.17, 8.33, 12.50, 16.67% → expected **7.083%** (or expected price 128.50 → 7.0833%); σ^2 34.902, σ **5.908%**. **TYK P2:** MP ₹ 106 cum ₹ 6 debenture → net cost ₹ 100; total returns 124/112/100 → net 24/12/0 → expected **₹ 12 (12%)**, σ = √72 = **8.49%**; 10% debenture interest vs 12% market — accept buy-back unless maturity is short (up to 1 year).

4.3.3 Measurement of Systematic Risk (Beta)

→ Systematic risk is measured by **Beta (β)** = security's variability relative to the market index. Inputs: historical returns of the security and of the market index. Two methods: correlation and regression.

Formula	Terms / logic	SM / TYK figures
4.3.3.1 Correlation method: $\beta_i = r_{im} \cdot \sigma_i \div \sigma_m$	r_{im} correlation of stock with market; σ_i, σ_m SDs. *App-written derivation:* since Cov = $r_{im}\sigma_i\sigma_m$, β = Cov ÷ σ_m^2 (form used in TYK P10, P12, P17).	TYK P14: A 20 × 0.60/15 = **0.80**; B 18 × 0.95/15 = **1.14**; C 12 × 0.75/15 = **0.60**.
4.3.3.2 Regression: $Y = \alpha + \beta X$; $\beta = [n\Sigma XY - (\Sigma X)(\Sigma Y)] \div [n\Sigma X^2 - (\Sigma X)^2]$; $\alpha = \bar{Y} - \beta\bar{X}$	Y = security return (dependent), X = market return (independent). β = change in Y per unit change in X; α = Y when X is zero. ICAI answers (Illus 6, TYK P11) also use $\beta = (\Sigma XY - n\bar{X}\bar{Y}) \div (\Sigma X^2 - n(\bar{X})^2)$, and TYK P10/P12 use β = Cov ÷ σ_m^2.	TYK P12: Σ dev products 357, Σ dev^2 M 706 → Cov 35.70, σ_m^2 70.60 → β **0.505** (alt 0.506); α = 15 − 0.505 × 12 = **8.94%** → line 8.94 + 0.505 R_M.

→ **Positive β**: moves with market; **negative β**: opposite to market; **zero β**: independent of market.

→ **β = 1**: risk as market; **β > 1**: above-average risk (β 2: market +6% → stock +12%, −6% → −12%); **β < 1**: below-market risk. Beta is a **historical** measure — assumes the relationship continues.

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TYK P9 (probabilities): E(F) **17%**, σ_F 11.87; E(P) **14%**, Var 264, σ 16.25; Cov(F,P) **−168** → β = −168/264 = **−0.636**.

TYK P10: A: Cov 4.83/3 = 1.61, σ_m^2 4.67/3 = 1.557 → β_A **1.03**; B: Cov 0.78 → β_B **0.50**.

4.4 Portfolio Analysis — 4.4.1 Portfolio Return

Formula	Terms / logic	SM example figures
$\bar{r}_p = \Sigma x_i \bar{r}_i$	x_i proportion of funds in security i; $\bar{r}_i$ its expected return. Simple weighted average.	A 16%, B 22%, 40:60 → 0.40 × 16 + 0.60 × 22 = **19.6%**.

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TYK P32 (X Co., M Ltd & N Ltd): 2009-10: M (10 + 20)/200 = 15%, N (3 − 10)/300 = −2.33%; weights 57:43 (cost) → **7.55%**. 2010-11: expected prices M 253, N 312; yields 53 and 25.5 on 220 and 290 → 24.09% and 8.79%; weights 60.3:39.7 (31.3.10 values) → **18.02%**. Risk in ₹: σ_M **21** (σ^2 441), σ_N **14** (σ^2 196); in %: σ_M **9.54%**, σ_N **4.83%** → M more risky.

4.4.2 Portfolio Risk

Asked: Sep 2025 Q4(a) 6m (two-stock risk, 60:40 reallocation)

→ Portfolio variance = weighted variances of constituents **+** weighted covariances among pairs. **Not** a simple weighted average of variances.

→ **Covariance**: how returns of two securities move relative to each other — same direction consistently (+), opposite (−), independent (close to 0).

Formula	Terms / logic	SM example figures
$Cov_{AB} = \Sigma[(R_A - \bar{R}_A)(R_B - \bar{R}_B)] \div N$ (with probabilities: $\Sigma p \times dev_A \times dev_B$)	Product of each period's deviations from the means, averaged over N observations.	4 years, $\bar{R}_x$ 15, $\bar{R}_y$ 13: Σ products −42 → Cov = −42/4 = **−10.5**.
$r_{AB} = Cov_{AB} \div (\sigma_A \sigma_B)$; $Cov_{AB} = \sigma_A \sigma_B r_{AB}$	Correlation ranges −1 to +1: −1 perfect negative, +1 perfect positive, 0 independent.	—
$\sigma_p^2 = x_1^2\sigma_1^2 + x_2^2\sigma_2^2 + 2x_1x_2 r_{12}\sigma_1\sigma_2$	$x_1 + x_2 = 1$. Last term = $2x_1x_2Cov_{12}$. $\sigma_p = \sqrt{\sigma_p^2}$.	A: r 20%, σ 50%; B: r 25%, σ 30%; r_{AB} −0.60; 40:60 → R_p **23%**; σ_p^2 = 400 + 324 − 432 = **292**; σ_p **17.09%**.

→ Return and risk depend on (a) returns, risks and covariances of securities — **parametric**, no control; (b) **proportions** — the investor's **choice** variable.

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TYK P3 (B σ 10%, ER 12%; D σ 18%, ER 20%; r 0.15): (i) 100% B: 12%, σ 10%; (ii) 50:50: 16%, σ^2 25 + 81 + 13.5 = 119.50, σ **10.93%**; (iii) 75:25: 14%, σ^2 56.25 + 20.25 + 10.125 = 86.625, σ **9.31%**; (iv) 100% D: 20%, 18%. Return → (iv) best; risk → (iii) best. **TYK P4:** E(A) 13%, E(B) 15%, 40:60 → **14.2%**; $\sigma_A = \sigma_B$ = 3%; Cov 9; r = 9/(3 × 3) = 1; $\sigma_p = \sqrt{(1.44 + 3.24 + 4.32)}$ = **3%**. **TYK P6 (10 years):** $\bar{R}_1$ 14.8, $\bar{R}_2$ 21; Σ products −8 → Cov **−0.8**; $\sigma_1 = \sqrt{20.76}$ = 4.56, $\sigma_2 = \sqrt{8.40}$ = 2.90 → r = −0.8/13.22 = **−0.0605**.

4.4.3 Reduction or Dilution of Portfolio Risk through Diversification

→ Diversification = combining more than one security to reduce total risk by eliminating/mitigating unsystematic risk **without sacrificing return**. 17.09% above < 50% and 30%. Possible even when uncorrelated; **no reduction at r = +1**.

Case	Formula	SM figures (σ_A 40, σ_B 25, 0.4:0.6)
4.4.3.1 r = +1	$\sigma_p = x_1\sigma_1 + x_2\sigma_2$ — weighted average of SDs; only **risk averaging**, no risk reduction.	0.4 × 40 + 0.6 × 25 = **31**; 0.75:0.25 → **36.25**. Lies between 40 and 25.
4.4.3.2 r = −1	$\sigma_p = x_1\sigma_1 - x_2\sigma_2$ (absolute) — can be made **zero**.	0.4 × 40 − 0.6 × 25 = **1**; 0.3846:0.6154 → **0** (risk-free portfolio of two risky assets). Rare in real life.
4.4.3.3 r = 0	$\sigma_p = \sqrt{(x_1^2\sigma_1^2 + x_2^2\sigma_2^2)}$ — covariance term drops.	√(256 + 225) = **21.93** — below both SDs.

r	1.00	0.60	0	−0.60	−1.00
σ_p	31	27.73	21.93	13.89	1.00

→ Diversification reduces risk in all cases **except r = +1**; as r falls from +1 to −1, σ_p falls; reduction greater with negative correlation.

Note (app-written): 4.4.3 says "taking the above example" but uses σ 40% and 25%, not the 50% and 30% of the 4.4.2 example (SM p. 23 vs p. 25). The table above uses ICAI's 40/25.

Formula (minimum-risk weights, as printed in TYK P7 answer)	Terms / logic	Figures
$X_{ABC} = (\sigma_X^2 - r_{AX}\sigma_A\sigma_X) \div (\sigma_A^2 + \sigma_X^2 - 2r_{AX}\sigma_A\sigma_X)$ or $= (\sigma_X^2 - Cov._{AX}) \div (\sigma_A^2 + \sigma_X^2 - 2Cov._{AX})$; $XYZ = 1 - X_{ABC}$	σ_X SD of XYZ; σ_A SD of ABC; r_{AX} correlation; $Cov._{AX}$ covariance. Numerator uses the **other** security's variance. *App-written:* with $r_{AX} = -1$ it reduces to $\sigma_X \div (\sigma_A + \sigma_X)$.	*App-written check on 4.4.3.2:* 25/(40 + 25) = 0.3846 (SM's zero-risk weight).

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TYK P7 (ABC, XYZ, ₹ 1,00,000): E(ABC) **12.55%**, σ^2 167.75, σ **12.95%**; E(XYZ) **12.1%**, σ^2 126.98, σ **11.27%**; Cov **−144.25**. 50:50 → σ^2 = 41.9375 + 31.745 − 72.125 = 1.5575, σ **1.25%**, return **12.325%**. Minimum risk: x_{ABC} = (126.98 + 144.25)/(167.75 + 126.98 + 288.5) = 271.23/583.23 = **46%**, XYZ **54%**.

4.4.4 Portfolio with more than two securities

→ Benefits increase as more securities with less than perfect positive correlation are added; σ_p keeps falling. In reality securities are positively (but not perfectly) correlated → some reduction, not elimination; effect **exhausted rapidly** — most reduction by **25 or 30 securities**; benefits of diversification limited in practice.

4.4.5 Return and Risk of Portfolio with more than two securities

Formula	Terms / logic	SM example figures
$\bar{r}_p = \Sigma\, x_i\, \bar{r}_i$	Same formula for any n.	P 11/0.3, Q 16/0.2, R 22/0.1, S 20/0.4 → **16.7%**.
$\sigma_p^2 = \Sigma_i \Sigma_j\, x_i\, x_j\, r_{ij}\, \sigma_i\, \sigma_j = \Sigma_i \Sigma_j\, x_i\, x_j\, \sigma_{ij}$	**Variance-covariance matrix**: each cell = weight (row) × weight (column) × r × σ × σ; r_{ii} = 1 on the diagonal (gives variance). Sum all n × n cells. *App-written expansion for three assets:* $x_X^2\sigma_X^2 + x_Y^2\sigma_Y^2 + x_Z^2\sigma_Z^2 + 2x_Xx_YCov_{XY} + 2x_Xx_ZCov_{XZ} + 2x_Yx_ZCov_{YZ}$.	X 0.25/σ16, Y 0.35/σ7, Z 0.40/σ9; r XY 0.7, XZ 0.3, YZ 0.4 → 16 + 6.86 + 4.32 + 6.86 + 6.0025 + 3.528 + 4.32 + 3.528 + 12.96 = **64.3785**; σ_p **8.0236**.

→ New portfolios come from changing the securities or their proportions; determining each portfolio's return and risk = **portfolio analysis**, the first step in portfolio management.

5. Markowitz Model of Risk-Return Optimization

→ Unlike CAPM, the optimal portfolio is **investor specific**. Two stages: (1) find **mean-variance efficient** portfolios; (2) select one.

→ Investors dislike risk; trade-off between risk and return; SD measures total risk; diversification removes a good deal of it.

5.1 Assumptions of the Model

→ **Return** adequately summarises the outcome; investors visualise a **probability distribution** of returns.

→ Risk estimates proportional to **variance**; decisions on two criteria only — **expected return and variance**.

→ Investors **rational and risk averse** (between equal returns, prefer smaller risk).

→ **Yield** in different scenarios is the measure of return; SD = deviation of yields from expected value.

5.2 Efficient Frontier

→ Built on the **Mean-Variance Dominance Principle**: a portfolio is dominated by another with higher (or same) return for same (or lower) risk. Diversification (Markowitz) = combining assets **less than perfectly positively correlated** to reduce risk without sacrificing return.

→ If not efficient, the investor may: increase return without more risk; reduce risk without lower return; or a combination — by **switching to a portfolio on the frontier**.

→ **Efficient portfolio**: highest return for identical risk and lowest risk for identical return. Figures: of A–F, **B, C, D** lie on the frontier; feasible set = shaded area; any frontier point dominates points to its right/below (B vs E: same return, B has less risk). **A** = least risk; **D** = highest return with highest risk.

→ **Optimal portfolio** = where the investor's **indifference curve touches the efficient frontier**; differs from investor to investor.

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TYK P5 (dominance): A (8%, σ 4) dominates B and F (same return, more risk) and D (same risk 4, return 4%) → select **A, C & E**. Perfect correlation, 75% A + 25% C: risk (3 × 4 + 1 × 12)/4 = **6%**, return **9%**; E gives 9% at risk **5** → **E preferable**. **TYK P31 (minimum variance set, short sales allowed):** ₹ 5,000 in X + ₹ 3,000 in Y → A 2,100, B 3,500, C 2,400 → weights **0.26, 0.44, 0.30**. Critical line $W_B = a + bW_A$: 0.40 = a + 0.30b, 0.50 = a + 0.20b → $\mathbf{W_B = 0.70 - W_A}$; ₹ 4,000 in A ($W_A$ 0.50) → W_B 0.20 (**₹ 1,600**), W_C 0.30 (**₹ 2,400**).

6. Capital Market Theory

→ B (tangent point on the frontier) = **market portfolio**; tangent line from $\mathbf{R_f}$ through B to Z = **Capital Market Line (CML)** — combinations of market portfolio with borrowing or lending.

→ R_f to B = **lending portfolios** (part risk-free, part market); B to Z = **borrowing portfolios** (borrow and invest in the market portfolio).

Formula	Terms / logic
Slope of CML = $(R_M - R_f) \div \sigma_M$	Reward per unit of (total) risk. R_M market return; R_f risk-free; σ_M SD of market.
$E(R) = R_f + [(R_M - R_f) \div \sigma_M] \times \sigma_P$	Expected return of an efficient portfolio = R_f + slope × its SD σ_P. CML uses **σ**; SML (CAPM) uses **β**.

7. Single Index Model (Sharpe Index Model)

Asked: Sep 2025 Q4(a) 6m

→ Co-movement between stocks is due only to movement in the **market index**; no effects beyond the market (no fundamental factor effects) explain co-movement. β and α estimated by regression.

Formula	Terms / logic	SM / TYK figures
$R_i = \alpha_i + \beta_i R_m + \in_i$	α_i intercept; β_i slope (sensitivity; β 1.5 → +1.5% per +1% market); R_m index return; $\in_i$ error term (unsystematic). Return = market part + part independent of market. $\alpha + \beta R_m$ = **characteristic line**.	TYK P11: $\beta = (508 - 6 \times 5.75 \times 6.33)/(439.25 - 6 \times 5.75^2) = 289.615/240.875 =$ **1.202**; $\alpha = 6.33 - 1.202 \times 5.75 =$ **−0.58** → line $-0.58 + 1.202\ R_m$.
$\sigma_i^2 = \beta_i^2\sigma_m^2 + \in_i^2$	Total variance = **systematic** (explained by index) + **residual/unsystematic** (unexplained).	TYK P11: σ_m^2 240.86/6 = 40.14; σ_A^2 497.34/6 = 82.89; systematic $1.202^2 \times 40.14 =$ **57.99**; unsystematic **24.90**.
Systematic = $\beta_i^2 \times \sigma_m^2$; Unsystematic $\in_i^2 = \sigma_i^2 -$ Systematic risk = Variance of Security Return $(1 - r^2)$ (Illus 1)	r^2 (coefficient of determination) = % of variation explained by the index.	Illus 1 below.
$\sigma_{ij} = \beta_i\beta_j\sigma_m^2$	Covariance between two stocks comes only through the market.	TYK P14: Cov A,B = 0.80 × 1.14 × 225 = **205.2**; A,C **108**; B,C **153.9**.
$\sigma_p^2 = [(\Sigma X_i\beta_i)^2\sigma_m^2] + [\Sigma X_i^2\in_i^2]$	First part = $\beta_p^2\sigma_m^2$ (portfolio systematic); second = weighted residual variances (weights **squared**). X_i = portion of stock i; $\in_i^2$ = variation not related to the index.	Illus 1: **3.5876**.
$R_P = \Sigma\ x_i(\alpha_i + \beta_i R_m)$; $\alpha_P = \Sigma\ x_i\alpha_i$; $\beta_P = \Sigma\ x_i\beta_i$	Portfolio return, alpha and beta are weighted averages using investment proportions.	—

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Illus 1 (X, Y, Sensex): σ_m^2 2.25. X: β 0.71, variance 6.30, r^2 0.18 → systematic $(0.71)^2 \times 2.25 =$ **1.134**; unsystematic 6.30 − 1.134 = **5.166** (= 6.30 × 0.82). Y: β 0.685, variance 5.86 → systematic **1.056**, unsystematic **4.804**. Equal weights: $[(0.355 + 0.3425)^2 \times 2.25] + [(0.5)^2(5.166) + (0.5)^2(4.804)] = 1.0946 + 2.493 =$ **3.5876**.

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TYK P14 (equal weights): matrix diagonals 400, 324, 144 (σ^2) → total variance **200.244**; $\beta_p = (0.80 + 1.14 + 0.60)/3 =$ **0.8467**; systematic $(0.8467)^2(15)^2 =$ **161.302**; unsystematic **38.942**. **TYK P34:** β_p **0.66**; σ_m^2 0.01 → residual A 0.0134, B 0.0225, C 0.0879; SIM variance 0.004356 + 0.014072 = **0.018428**; Markowitz (9-cell matrix with covariances) **0.0363**. **TYK P37:** β_p **1.295**; $\sigma_p = [543.36 + 3.0625 + 10.89 + 0.5625 + 3.24]^{1/2} = \sqrt{561.115} =$ **23.69%** (ICAI alternative: 25.03%).

Note (app-written): TYK P37's "alternative answer" (SM p. 125) weights each security's total variance by w (not w^2) and adds no covariance, so its 25.03% is not the single-index portfolio risk. The formula above gives ICAI's main answer, 23.69%.

8. Capital Asset Pricing Model (CAPM)

Asked: Jan 2026 MCQ CS-III 7-9 6m

→ Portfolio risk = unsystematic (averted by diversification) + systematic (market-related, measured by **β**). Only **non-diversifiable** risk is priced; market $\beta = 1.0$; portfolio $\beta = \Sigma$ proportion × β.

→ Three aspects: market not concerned with diversifiable risk; not concerned with investor holding a diversified portfolio; **compensation only for non-diversifiable risk**.

Assumptions (TYK T3):

→ **Efficient market** (full information, prices reflect intrinsic value; Hampton — CAPM usable for capital budgeting); **rational investment goals**; **risk aversion**.

→ Assets **divisible and liquid**; **borrow freely at risk-less rate**; **no brokerage, taxes or transaction cost**; **no bankruptcy**; capital market in **equilibrium**.

→ Required return = minimum expected return for investors to buy and hold the asset.

Formula	Terms / logic	SM figures
$R_j = R_f + \beta(R_m - R_f)$	R_f risk-free return; $(R_m - R_f)$ **market risk premium** (reward for market risk); β × premium = stock's risk premium (β tells how much riskier than market).	R_f 5%, R_m 15% → premium 10%; 30% riskier → 10% + 3% = 13% → 5% + 1.3 × 10% = **18%**. Premium 6%, β 1.2 → **7.2%**.
SML: required return (Y-axis) vs β (X-axis); intercept R_f, slope $(R_m - R_f)$	Relationship between **systematic risk** and return; higher β → higher premium.	—
$\beta_p = \Sigma w_i\beta_i$ (risk-free β = 0, market β = 1)	Mixing T-bills and market changes β linearly.	Illus 4.

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Illus 2: β 1.40, R_m 14%, R_f 10% → 10% + 5.6% = **15.6%**; premium +2.50 points → R_m 16.5% → 10% + 1.40 × 6.5% = **19.10%**. **Illus 3:** premium 10%, β 0 / 0.25 / 0.42 / 1.00 / 1.67 → **0% / 2.50% / 4.20% / 10.00% / 16.70%**. **Illus 4:** R_f 5%, R_m 13% → premium **8%**; T-bill:market 100:0 β 0 → 5%; 70:30 β 0.3 → **7.40%**; 30:70 β 0.7 → **10.60%**; 0:100 β 1 → 13%.

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TYK P13: β aggressive (40 − 4)/(25 − 7) = **2**; defensive (18 − 9)/(25 − 7) = **0.50**; expected returns **22%** and **13.5%**; R_m 16% → SML: **7.5% + β_i 8.5%**; alphas ($R_s = \alpha + \beta R_m$): α_A **−10%**, α_D **5.5%**. **TYK P16:** premium 7% → returns 10.15/9.45/15.05/19.95% → ₹ 1,45,933 on ₹ 9,05,000 = **16.13%**; β **1.30** (weighted 1.3035 → 16.1245%). **TYK P19:** R_f = 7/140 = 5%; 15% = 5% + β(10%) → β 1; 1 = 225/σ_m^2 → σ_m **15%**; 1 = σ_X × 0.75/15 → σ_X **20%**. **TYK P20 (two correctly priced shares):** 19.8 = R_f + 1.2P, 17.1 = R_f + 0.9P → P = 2.7/0.3 = 9 → R_m **18%**, R_f **9%**; SML = 9% + β × 9%. **TYK P21:** B better (higher return, lower risk); 70:30 → 22.6%, σ^2 0.1374, σ **37%** (37.06% also correct); $R_m - R_f$ = −2/−0.38 = 5.26% → R_f **17.48%**, R_m **22.74%**; β_p **0.974**.

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TYK P23: 20% each → β **1.22**; 15% A, 30% B, 15% C, 30% D, 10% E → β **1.335**; at 12% for β 1 → **14.64%** and **16.02%**. **TYK P25 (₹ 520 lakh):** CAPM returns 9 / 10 / 9.60 / 10.40 / 11% → portfolio **10.208%** (avg β 1.104); "Better" has β 1 = Nifty's 10% → replacement makes no difference. **TYK P36 (beta adjustment):** market value ₹ 12,00,000, β_p **1.108**; to 0.8 → 0.8/1.108 = 72.2% → total ₹ 16,62,050 → buy risk-free **₹ 4,62,050** (alternative: 13,30,000/(12,00,000 + x) = 0.8 → **₹ 4,62,500**); to 1.2 → 108.30% → ₹ 11,08,030 → divest **₹ 91,970** (alternative **₹ 91,667**). **TYK P39:** β_{asset} = 1.5 × 60/100 = **0.9** (debt riskless); cost of capital 8% + 0.9 × 10 = **17%** (or WACC 23% × 3/5 + 8% × 2/5 = 17%).

Note (app-written): TYK P36's two methods differ only by rounding of 72.2% and 108.30% (SM p. 123–125); the direct equation gives ₹ 4,62,500 and ₹ 91,667. TYK P23(iii) takes return = 12% × β with no risk-free rate given (SM p. 111).

8.1 Risk free Rate of Return

Asked: Jan 2026 MCQ (R_f from average portfolio return)

→ CAPM has **only one** risk-free rate. Returns include both **dividend and capital appreciation** — factor both into β and required return.

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Illus 5 (Pearl Ltd): T-bill lowest bid 4.3% (conservative), 4.6% (aggressive) → moderate average **4.45%**; 15.50% = 4.45% + β(12% − 4.45%) → β = 11.05/7.55 = **1.464**. **Illus 6 (Jaykay):** returns = $[D_1 + (P_1 - P_0)]/P_0$: 25.62%, 20.07%, 17.05%; market = index appreciation + dividend yield: 12.62%, 21.79%, 5.32%; β = (851.36 − 830.55)/(662.36 − 525.89) = 20.81/136.47 = **0.15** → less affected by market. **TYK P8 (Krishna):** returns 12.24/32.41/16.13% (avg 20.26); market 10.81/16.33/16.79% (avg 14.64); β **1.897**; R_f 6% → expected 15.12% (Sell), 25.60% (Buy), 26.47% (Sell).

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TYK P26: dividends 7,025 + capital gain 5,500 on cost 75,000 → R_m **16.7%**; average β (0.6 + 0.8 + 0.6 + 0.01)/4 = 0.50 → 15.7 = R_f + 0.50(16.7 − R_f) → R_f **14.7%**; Gold 15.90%, Silver 16.30%, Bronze 15.90%, GOI 14.72%. **TYK P27:** R_m = (146 + 145)/1,105 = **26.33%**; Epsilon 23.86%, Sigma 22.63%, Omega 20.17%, GOI 14.12% → average **20.20%**. **TYK P28:** R_m = (5,800 + 5,000)/68,000 = **15.88%**; X 15.70%, Y 15.62%, Z 15.44%, PSU 15.18% → **15.49%**.

Note (app-written): TYK P26's alternative (SM p. 113–114) computes β 0.387 and R_f 5.83% from the GOI bond, then stops at 15.7% = 5.83% + 0.387(R_m − 5.83%) without solving. Use the main answer.

8.2 Under Valued and Over Valued Stocks

Compare	Valuation	Decision
CAPM required return < actual/expected return (positive alpha) or actual price < CAPM price	Undervalued	**Buy**
CAPM required return > actual return (negative alpha) or actual price > CAPM price	Overvalued	**Sell**
Equal	Correctly valued	**Hold**

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Illus 7: 10 + 1.2(15 − 10) = **16%** < projected 18% → undervalued → **Buy**. **Illus 8:** R_f 9%, R_m 14%: A required **17.50%** vs 18% → under-valued, Buy; B **12.00%** vs 11% → over-valued, Sell; C **15.00%** vs 15% → correctly valued, Hold. **TYK P22 (XYZ):** market value 11,00,500, dividends 2,46,282 → return **22.38%**; β_p **1.46** vs implicit β (0.2238 − 0.11)/0.08 = **1.42** → marginally riskier than market; CAPM I 20.28%, II 29.24%, III 18.20%, IV 23% → I, II, III yields below → over-priced, sell; IV yield 26% above → increase. **TYK P33:** E(A) 11.5%, E(B) 10.1%, E(M) 10.2%, Var M 78.96, Cov AM 106.20, BM 106.68 → β_A **1.345**, β_B **1.351**; required (R_f 11%) 9.924% and 9.92% → α_A **1.576%**, α_B **0.18%** → underpriced → make fresh investment.

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CAPM + dividend growth model ($P_0 = D_0(1 + g) \div (K_e - g)$): TYK P18 (Amal): 10 + 1.2 × 5 = **16%**; P_0 = 3.36/0.04 = ₹ **84**. **TYK P29 (Fatlass, MP ₹ 25):** existing K_e 20.4% → P_0 2.10/0.154 = ₹ **13.63**; revised 15% → 2.18/0.06 = ₹ **36.33** → **hold** (expect appreciation). **TYK P30 (X Ltd, MP ₹ 40):** 18.5% → 3.27/0.095 = ₹ **34.42**; 15.60% → 2.675/0.086 = ₹ **31.10** → **sell**.

8.3 Advantages and Limitations of CAPM

Advantages	Limitations
Risk adjusted return — *reasonable basis for required return; usable as risk-adjusted discount rate in capital budgeting.* ***No dividend company*** — *computes cost of equity of a company that pays no dividend.*	***Reliability of beta*** — *reliable β may not exist for many firms; beta's shortcomings apply to CAPM.* ***Other risks*** — *ignores unsystematic risk.* ***Information available*** — *hard to get R_f (multiple rates) and expected R_m (volatile).*

9. Arbitrage Pricing Theory Model (APT)

Asked: May 2026 Q2(a) 6m

→ *Alternative to CAPM: uses the asset's expected return and the* ***risk premia of several macro-economic factors.*** *Steps: (a) risk premium for each factor × its beta (sensitivity); (b) add R_f.*

→ *CAPM = one factor (market); APT needs no particular factor.* ***Stephen Ross*** *factors:* ***inflation, level of industrial production, risk premium, term structure of interest rates.*** *No surprise in factors → actual = expected.*

Formula	Terms / logic	SM figures
$E(R_i) = R_f + \lambda_1\beta_1 + \lambda_2\beta_2 + \ldots + \lambda_n\beta_n$	λ_n = factor price / risk premium of factor n; β_n = sensitivity to factor n. Expected return in equilibrium.	Illus 9 below.
With surprises: $E(R) = R_f + \beta_1(EV_1 - AV_1) + \ldots + \beta_n(EV_n - AV_n)$	(EV − AV) = surprise in factor value; β = sensitivity to that macro factor. (ICAI's TYK P38 uses actual − expected.)	TYK P38.
b prop. 1 = Σ (position in security ÷ own funds) × its Factor 1 sensitivity (same for b prop. 2) — as in TYK P15	Position ÷ own funds; short sale and borrowing are negative weights; risk-free asset has sensitivity 0.	TYK P15.

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Illus 9: *8% + 1.3 × 4% + 0.3 × 1% + 0.2 × (−4%) = 8 + 5.2 + 0.3 − 0.8 =* ***12.7%.*** ***TYK P38 (Tamarind):*** *(actual − expected) × β: 0 + 2.63 + 1.63 + 3.40 + 0.50 = 8.16 → 9.25 + 8.16 =* ***17.41%.*** ***TYK P15:*** *(i) +1.50 in A, −0.5 in B → b_1 1.50 × 0.80 − 0.50 × 1.50 =* ***0.45****, b_2* ***0.30****; (ii) borrow ₹ 1,00,000: A 3, B −1, R_f −1 →* ***0.90*** *and* ***0.60****; (iii) 15 = 10 + $0.80\lambda_1 + 0.60\lambda_2$, 20 = 10 + $1.50\lambda_1 + 1.20\lambda_2$ → λ_1 0, λ_2 = 5/0.60 =* ***8.33%.***

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TYK P40 (Nirmal Kumar): *(a) APT: small cap growth 5.9925%, small cap value 8.8525%, large cap growth 8.478%, large cap value 7.535% → market index* ***7.7526%*** *(Method II 4.5 + 6.85 − 7.3675 + 3.77 = 7.7525%); (b) CAPM: 9.98, 10.665, 12.48, 10.3225 →* ***11.33%****; (c) target β 1 with small cap value (0.90) and large cap growth (1.165): X_1 + X_2 = 1, $0.90X_1 + 1.165X_2 = 1$ → X_1 = 0.165/0.265 =* ***62.3%****, X_2* ***37.7%****.*

Note (app-written): in TYK P40(b) ICAI takes 12.45 for large cap growth in the total, while its own line above gives 12.48 (SM p. 127); with 12.48 the total is about 11.35%. In TYK P38 the rounded products add to 8.16; the unrounded total is 8.15 (17.40%).

10. Portfolio Evaluation Methods (10.1–10.3)

Formula	Terms / logic	SM example figures
10.1 Sharpe $S = (R_i - R_f) \div \sigma_i$	Risk premium per unit of **total risk** (SD). Higher = better risk-adjusted return. Use for overall portfolio vs another portfolio or index.	Index fund 11%, T-bill 6%, σ 20% → **0.25**. A (30%, σ 11) **2**; B (25%, σ 5) **3.4** → B better despite lower return.
10.2 Treynor $T = (R_i - R_f) \div \beta_i$	Risk premium per unit of **systematic risk**; unsystematic assumed diversified.	β_A 1.5 → **14.67%**; β_B 1.1 → **15.45%** — same ranking as Sharpe.
10.3 Jensen alpha = actual return − CAPM expected return $[R_f + \beta_p(R_m - R_f)]$	Ability of active management to add return above reward for market risk. Meaningful only for portfolios with **similar betas**. +α = manager doing well; −α = not doing well.	A: 12%, β 0.7, R_f 9%, R_m 12% → 0.09 + 0.021 = 0.111 → α = **0.009**.

Sharpe vs Treynor	Sharpe	Treynor
Risk used	SD — total (systematic + unsystematic) volatility	Beta — systematic only ("momentum" built in)
Better for	**Sector-specific** / not fully diversified funds	**Diversified equity** funds
Rankings	Usually similar (most portfolios fully diversified)	Usually similar (most portfolios fully diversified)

→ Alpha: compare "apples to apples" (sector fund vs sector fund); alphas from different rating services not comparable; short-term alphas not valid — minimum **one year**, **three years** preferable.

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TYK P17 (MFX, MFY, Mix): variances = diagonal: **4.800, 4.250, 3.100**; weights 0.60/0.40 → return **14.60%**; β_X 3.370/3.100 = **1.087**, β_Y 2.800/3.100 = **0.903**, β_p **1.013**; σ_p^2 = 0.36 × 4.8 + 0.16 × 4.25 + 2 × 0.6 × 0.4 × 4.3 = **4.472**, σ_p **2.115**; CAPM X 12.17%, Y 11.81%, portfolio 12.03%; systematic 3.663 / 2.528 / 3.181, unsystematic 1.137 / 1.722 / 1.291; Sharpe **2.282 / 1.94 / 2.175**; Treynor **4.60 / 4.43 / 4.54**; alpha **2.83% / 2.19% / 2.57%**.

Note (app-written): in TYK P17 the Sharpe ratios print the variance (4.800, 4.250) as denominator, but the answers use SD: 5/√4.8 = 2.282, 4/√4.25 = 1.94 (SM p. 106).

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TYK P24 (Economy A): (1 − w)10 + w15 = 10.5 → w **10%**; $\sigma^2 = (0.9)^2(0.16)^2 + (0.1)^2(0.30)^2 + 2(0.9)(0.1)(0.16)(0.30)(0.30) = 0.02423$ → σ **15.6%**; Sharpe (10 − 3)/16 = **0.437** → (10.5 − 3)/15.6 = **0.481** (improves ~0.04). **TYK P41 (R_f 6%):** Sharpe A 1.285, B 1.20, C 1.60, D 1.00, E 1.11 → ranks C, A, B, E, D; Treynor A 7.2, B 16, C 5.71, D 6.12, E 6.67 → ranks B, A, E, D, C. **TYK P42:** $\beta = \sigma_p r/\sigma_m$: A 1.75, B 0.90, C 0.65, D 1.25, E 0.90; required 17.75 / 13.50 / 12.25 / 15.25 / 13.50; Sharpe 4.00 / 3.00 / 7.50 / 4.25 / 4.50 (C first); Treynor 5.71 / 6.67 / 9.23 / 6.80 / 9.00 (C first); Jensen 1.25 / 1.50 / 2.75 / 2.25 / 3.60 (E first).

11. Sharpe's Optimal Portfolio

Asked: Jan 2026 Q5(a) 8m

Asked: May 2026 Q4(a) 6m

→ Simplified variant of Markowitz — far less data and computation. Based on **excess return over R_f to beta**; ranking plus a unique **cut-off point C***.

Step / formula	Terms / logic
(a) $(R_i - R_f) \div \beta_i$ for each stock (Treynor's ratio); (b) rank highest to lowest	Excess return per unit of systematic risk.
(c) $C_i = \sigma_m^2 \Sigma_{i=1}^{N} [(R_i - R_f)\beta_i \div \sigma_{ei}^2] \div \{1 + \sigma_m^2 \Sigma_{i=1}^{N} [\beta_i^2 \div \sigma_{ei}^2]\}$	σ_m^2 = market index variance; σ_{ei}^2 = stock's unsystematic (residual) variance. Sums run from the top-ranked stock down to stock i (cumulative in ranked order, as in TYK P35).
(d) C^* = highest C_i	Select stocks whose $(R_i - R_f)/\beta_i$ is **above** C^*; reject those below. If a stock belongs to the optimal portfolio, all higher-ranked ones do too.
(e) $Z_i^0 = (\beta_i \div \sigma_{ei}^2) \times [(R_i - R_f)/\beta_i - C^*]$; $X_i^0 = Z_i^0 \div \Sigma_{j=1}^{N} Z_i$	Z_i gives relative investment; dividing by ΣZ makes weights total 1 (all funds invested).

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TYK P35 (Ramesh, σ_m^2 10, R_f 7%): excess return/β: A 5.33, B 2.5, C 1.2, D 2, E 0.83, F 4.67 → rank A, F, B, D, C, E. C_i: $C_A = 10 \times 0.30/[1 + 10 \times 0.056] =$ **1.923**; $C_F = 10 \times 0.65/[1 + 10 \times 0.131] =$ **2.814**; C_B 2.668; C_D 2.542; C_C 2.165; C_E 2.047 → **$C^* = 2.814$** → select A and F. $Z_A = (1.5/40)(5.33 - 2.814) =$ **0.09435**; $Z_F = (1.5/30)(4.67 - 2.814) =$ **0.0928** → A **50.41%**, F **49.59%**.

Note (app-written): the opening line of section 11 says ranking is "based on the Sharpe Ratio", but step (a) and TYK P35 rank by excess return to beta, i.e. Treynor's ratio (SM p. 53).

12. Formulation of Portfolio Strategy — 12.1 Active Portfolio Strategy (APS)

- → Followed by most professionals and aggressive investors to earn superior risk-adjusted return and beat the index; most Indian funds are active. Four principles:
- → **(a) Market timing** — depart from long-run asset mix on a forecast of market movement (business cycle, moving average, advance-decline analysis, econometric models); few succeed; forecasting right 75% of the time only breaks even after costs; Fisher Black — investor loses by being out of the market part of the time.
- → **(b) Sector rotation** — overweight sectors expected to do well (e.g. steel, pharma); for bonds, shift in quality, coupon, maturity (expect rate rise → long to medium/short term).
- → **(c) Security selection** — search for under-priced securities by fundamental/technical analysis; overweight attractive, underweight unattractive; bonds: highest YTM at given risk.
- → **(d) Specialised investment concept** — growth, neglected/out-of-favour, asset, technology, cyclical stocks. Helps focus on one's ability, avoid distraction, master a style; disadvantage — **may become obsolete**.

12.2 Passive Portfolio Strategy

- → Capital market fairly efficient. Two guidelines: (a) create a **well-diversified portfolio at a predetermined risk**; (b) **hold it relatively unchanged** unless inadequately diversified or inconsistent with risk-return preference.
- → **Index fund** — invests in the target index in the same weights (e.g. Nifty fund in the 50 Nifty companies); returns track the index; does **not** mitigate market risk.

12.3 Selection of Securities — 12.3.1 Bonds; 12.3.2 Equity

12.3.1 Bonds (fixed income) — factors	12.3.2 Equity — approaches
Yield to maturity (return if held to maturity); **risk of default** (credit rating, else debt-equity, interest coverage, earning power, industry prospects); **tax shield** (few offer now); **liquidity** (convert to cash at short notice).	**Technical** (price and volume data); **fundamental** (earnings, growth, risk → intrinsic value vs market price); **random selection** (market efficient, securities properly priced).

Market efficiency	Technical	Fundamental	Random selection
Inefficiency	Best	Poor	Poor
Weak form	Poor	Best	Poor
Semi-strong	Poor	Good	Fair
Strong form	Poor	Fair	Best

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TYK T1: factors affecting investment in fixed income securities → para 12.3 (12.3.1: YTM, risk of default, tax shield, liquidity).

13. Portfolio Revision and Rebalancing

→ Stock and bond values fluctuate, so the mix changes → rebalancing needed. Three policies (portfolio of stocks and bonds), different pay-offs in different markets.

Policy	Rule	SM example figures
(a) Buy and hold ("do nothing")	No rebalancing; set a **floor** and put an amount equal to it in bonds; portfolio value **linearly** related to stock value → **straight-line** pay-off. Suits investor whose risk tolerance rises with wealth/market but is zero below floor. Features: value positively and linearly dependent on stock; cannot fall below floor (bonds); better if initial stock % high and stock outperforms bonds.	Bonds ₹ 50,000 + equity ₹ 50,000 at ₹ 100; price ₹ 100 → ₹ 200 → portfolio **₹ 1,50,000**.
(b) Constant mix ("do something")	Keep stock at a **constant % of total**: when price falls by the trigger %, sell bonds and buy stock; when price rises, sell stock and buy bonds. Suits investor whose tolerance varies proportionally with wealth (holds equity at all levels). **Concave** pay-off; does well in **flat but fluctuating** market.	Text: 60:40, rebalance on 10% move. Table (50:50 on ₹ 1,00,000): price 80 → 40,000 + 50,000 = 90,000 → 45,000/45,000, bond to stock **5,000**; price 60 → 33,750 + 45,000 = 78,750 → 39,360/39,390, bond to stock **5,610**. Price 150 → 75,000 + 50,000 = 1,25,000 → 62,400/62,600, stock to bond **12,600**; price 200 → 83,200 + 62,600 = 1,45,800 → 72,800/73,000, stock to bond **10,400**.
(c) CPPI (Constant Proportion Portfolio Insurance)	Floor value in non-fluctuating assets (T-bills, bonds); **Target shares = Multiplier × (Portfolio value − Floor)**, multiplier > 1; (portfolio − floor) = **cushion**. Buy shares as prices rise, sell as they fall → **convex** pay-off; good in bull market, avoids losses in bear market; very poor in sharp reversals.	Mr A: wealth ₹ 10,00,000, floor ₹ 7,50,000, multiplier 2 → cushion 2,50,000 → shares **₹ 5,00,000**, bonds ₹ 5,00,000. Market 100 → 150: shares 7,50,000, portfolio 12,50,000, cushion 5,00,000 → target 10,00,000 → sell bonds, buy shares **₹ 2,50,000**. Market 100 → 80: shares 4,00,000, portfolio 9,00,000, cushion 1,50,000 → target 3,00,000 → sell shares **₹ 1,00,000**, invest in bonds.

Note (app-written): (1) Constant mix: the text sets 60:40 but the table rebalances 50:50, and ICAI's "after rebalancing" pairs are not exactly equal (39,360/39,390 for 78,750; 62,400/62,600 for 1,25,000; 72,800/73,000 for 1,45,800; exact halves 39,375, 62,500, 72,900) (SM p. 59–60). The switching amounts follow ICAI's rows. (2) CPPI Situation 1: SM says the cushion becomes ₹ 7,50,000, but 12,50,000 − 7,50,000 = ₹ 5,00,000, which is what gives ICAI's target of ₹ 10,00,000 (5,00,000 × 2) (SM p. 61).

(d) Comparative evaluation	Buy & Hold	Constant Mix	CPPI
Pay-off line	Straight	Concave	Convex
Down/Up markets	Definitely poor in down market	Not much in down market but relatively poor in up market	Good in down market and performs well in up market
Flat market	Between constant mix and CPPI	Tends to do well	Performs poorly

14. Asset Allocation Strategies

- → **Integrated** — examine capital market conditions and investor objectives/constraints; allocation that best serves needs with the market forecast.
- → **Strategic** — optimal mixes from historical returns, risk, covariances; adjusted periodically to restore **target allocation**.
- → **Tactical** — risk tolerance assumed **constant**; allocation changed on expectations about market conditions.
- → **Insured** — risk exposure adjusted with portfolio value; more wealth → more ability to take risk.

15. Fixed Income Portfolio — 15.1 Process; 15.2 Return

- → Like an equity portfolio but of bonds, debentures, money market instruments (**bond portfolio**).
- → **15.1 Five steps:** set objective → draft investment policy guidelines → select strategy (active/passive) → select securities and other assets → evaluate performance against benchmark.
- → **15.2 Return measures:** arithmetic average rate of return; time weighted rate of return; rupee weighted rate of return; annualised return.

15.3 Strategies — 15.3.1 Passive; 15.3.2 Active

15.3.1 Passive (securities fairly priced)	15.3.2 Active (to outperform)
Buy and hold — do nothing; may set duration equal to benchmark to control interest rate risk. **Indexation** — replicate a well-known bond index. **Immunization** — hybrid; popular with pension funds; avoids interest rate risk on fixed liabilities. **Matching cash flows** — buy zero coupon bonds to meet promised payments.	**(1) Forecasting returns and interest rates** — expect fall in rates → buy longer maturity. Short-term yield strategies: **bullet** (concentrate in one bond, e.g. 100% in 5-year bond — for a fixed future need); **barbell** (equal in short and long, e.g. 1-year and 10-year); **ladder** (equal amounts across maturities, e.g. 20% each in 1–5 years). Forecast models: inflation, past trends, multi-factor analysis (estimates only); **horizon analysis** — pick a holding period and predict the yield curve at its end. **(2) Bond swaps: pure yield pickup** (lower to higher yield, similar quality/maturity; bears interest rate risk); **substitution** (identical features, different price — temporary imbalance); **international spread** (yield spread between two sectors temporarily out of line); **tax swap** (sell at capital loss to set off gains, buy a similar bond). **(3) Interest rate swap** (see Interest Rate Risk Management chapter).

Note (app-written): 15.3.2(1) prints "if he/she expected a fall in interest then he/she should sell bonds with longer period"; read with the line before it, this should say a rise in interest rates (SM p. 64).

16. Alternative Investment Avenues

- → AIs = investments other than stock, bond and cash. **Features:** high fees; limited historical data; illiquidity; less transparency; extensive research required; leveraged buying.
- → Types: real estates, gold, private equity, REITs, hedge funds, ETFs, mutual funds, commodities, distressed securities (others covered in their chapters).

16.1 Real Estates (16.1.1 Valuation approaches; 16.1.2 Valuation)

→ Tangible: land, buildings, offices, warehouses, shops; country-specific laws reduce appeal of foreign property (rising with globalisation).

→ **16.1.1 Why valuation is complex:** inefficient market; illiquidity; only approximate comparison; high transaction cost; no organised market.

→ **16.1.2 Approaches: sales comparison** (like P/E multiplier — benchmark similar property); **income** (like perpetual debenture — perpetual net income ÷ required return); **cost** (replacement cost of building + land value, adjusted for location etc.); **discounted after-tax cash flow** (PV of inflows – investment, like NPV).

16.2 Gold

Asked: May 2026 MCQ CS-I 1-3 6m (SGB)

→ Jewellery suffers **making charges** on buying and selling back. Alternatives:

→ **Gold bars/coins** — but cost of physical storage.

→ **Sovereign Gold Bonds** — Government securities in **grams of gold**, issued by **RBI on behalf of GoI**; pay issue price in cash, redeemed in cash at market price of gold; periodical interest; no storage risk/cost, no making charges or purity issue; held in RBI books or demat.

→ **Gold ETFs** — traded on exchanges like stock; invest in gold bullion; transparent holdings; much lower expenses.

→ **E-gold** — from 2010, National Spot Exchange Ltd (NSEL); 1 unit = 1 gram, held in demat; backed by gold with custodian.

The May 2026 SGB MCQs also computed post-tax interest, real return, indexed cost and capital gains tax; the SM's 16.2 gives only the features above.

16.3 Distressed securities

→ Securities of companies in or near bankruptcy, bought cheap, to try to revive the company; suit investors who cannot participate in the market and want to avoid due diligence.

→ Profit: **long position in debt + short position in equity** — if condition improves, interest (priority) exceeds dividend on the short; if it deteriorates, both fall and the short gains.

→ **Risks:** liquidity risk; event risk (company-specific events); market risk (not important); human risk (judge's decision).

Exam steps by numerical type

Exam steps — expected return and SD: returns = $[D_1 + (P_1 - P_0)] \div P_0$ (net cost if cum-benefit) → $E(R) = \Sigma\ p \times R$ (or mean) → deviations → $\sigma^2 = \Sigma\ p \times dev^2$ (or ÷ n) → $\sigma = \sqrt{}$ (TYK P1, P2, P32).

Exam steps — two-asset portfolio and minimum risk: means → deviations → $Cov = \Sigma\ dev_A \times dev_B$ (× p or ÷ n) → $r_{AB} = Cov_{AB} \div \sigma_A\sigma_B$ → $\sigma_p^2 = x_1^2\sigma_1^2 + x_2^2\sigma_2^2 + 2x_1x_2r_{12}\sigma_1\sigma_2$ → min-risk $X_{ABC} = (\sigma_X^2 - Cov._{AX}) \div (\sigma_A^2 + \sigma_X^2 - 2Cov._{AX})$ (TYK P3, P4, P6, P7).

Exam steps — three-asset portfolio: write the 3 × 3 matrix (diagonal σ^2, off-diagonal $r\sigma\sigma$ or Cov) → multiply each cell by row weight × column weight → add all 9 → $\sqrt{}$ (4.4.5 example, TYK P34(iv)).

Exam steps — beta: regression $\beta = [n\Sigma XY - (\Sigma X)(\Sigma Y)] \div [n\Sigma X^2 - (\Sigma X)^2]$ with X = market, Y = security; or $\beta = Cov \div \sigma_m^2$; or $\beta = r_{im}\sigma_i \div \sigma_m$ → $\alpha = \bar{Y} - \beta\bar{X}$ → characteristic line $\alpha + \beta R_m$ (Illus 6, TYK P8–P12).

Exam steps — single index risk split: systematic = $\beta^2\sigma_m^2$ → unsystematic $\in_i^2 = \sigma_i^2 -$ systematic (or $\sigma_i^2(1 - r^2)$) → $\beta_P = \Sigma x_i\beta_i$ → $\sigma_p^2 = [(\Sigma X_i\beta_i)^2\sigma_m^2] + [\Sigma X_i^2\in_i^2]$; for a new mix (e.g. 60:40) only change w (Illus 1, TYK P14, P34, P37).

Exam steps — CAPM valuation: R_f (T-bill / bond yield = coupon ÷ price) and R_m (dividend + capital gain ÷ cost) → β → required $R_j = R_f + \beta(R_m - R_f)$ → compare with expected: required < expected → undervalued, buy; > → sell; = → hold; for price use $P_0 = D_1 \div (K_e - g)$ vs CMP (Illus 7, 8, TYK P18, P29, P30, P33).

Exam steps — R_f and R_m from data: two correctly priced securities → subtract the two CAPM equations → $(R_m - R_f)$ → back-substitute; portfolio average return with average β → solve R_f (TYK P20, P21, P26).

Exam steps — beta adjustment / target beta: $\beta_p = \Sigma$ (value ÷ total) × β → required total = (Σ value × β) ÷ target β → risk-free to add (or divest) = required total − current total; target β from two stocks: $w_1 + w_2 = 1$ and $w_1\beta_1 + w_2\beta_2$ = target (TYK P36, P40(c)).

Exam steps — APT: $R_f + \Sigma \lambda_k\beta_k$ (or Σ β × surprise) → compare with actual return (mispricing) → portfolio sensitivity = Σ (position ÷ own funds) × β, short and borrowing negative → for target sensitivities solve two simultaneous equations in the weights (Illus 9, TYK P15, P38, P40).

Exam steps — Sharpe / Treynor / Jensen ranking: $\beta = r\sigma_p \div \sigma_m$ if not given → Sharpe $(R_p - R_f) \div \sigma_p$ → Treynor $(R_p - R_f) \div \beta$ → Jensen $R_p - [R_f + \beta(R_m - R_f)]$ → rank highest first (TYK P17, P41, P42).

Exam steps — Sharpe optimal portfolio: $(R_i - R_f) \div \beta$ → rank descending → for each row $(R_i - R_f)\beta_i \div \sigma_{ei}^2$ and $\beta_i^2 \div \sigma_{ei}^2$, keep running totals of both → $C_i = \sigma_m^2 \times$ first total $\div (1 + \sigma_m^2 \times$ second total) → C^* = highest C_i (or given) → select ratios above C^* → $Z_i = (\beta_i \div \sigma_{ei}^2)(\text{ratio} - C^*)$ → $X_i = Z_i \div \Sigma Z$ (TYK P35).

Exam steps — rebalancing: CPPI: cushion = portfolio − floor → target shares = multiplier × cushion → buy/sell the gap against bonds. Constant mix: revalue shares at new price → total → restore target % → switch the difference. Buy and hold: value = bonds (floor) + shares × new price (13 examples).

⑥ Portfolio Management — what ICAI asked (Sep 2025, Jan 2026, May 2026)

38 marks *set across the last three attempts (10.7% of marks set). Question labels and marks are ICAI's; the one-line concept descriptions are app-written.*

Attempt	Question	Marks	Concept tested
Sep 2025	Q4(a)	6	Single-index risk split: systematic risk (beta squared x market variance) and unsystematic risk; two-stock portfolio risk; effect of a 60:40 reallocation (Numerical)
Jan 2026	MCQ CS-III 7-9	6	CAPM/SML from portfolio data: market return, risk-free rate from average portfolio return and betas, expected return of one share (MCQ)
Jan 2026	Q5(a)	8	Ranked table by Treynor ratio, cut-off point and securities in the optimal portfolio (single index model) (Numerical)
May 2026	MCQ CS-I 1-3	6	Sovereign Gold Bond (SGB): post-tax annual interest and real post-tax return; indexed cost and capital gain; net amount after capital gains tax (MCQ)
May 2026	Q2(a)	6	Two-factor APT: required return = Rf + sum of risk premium x beta, tested against actual return; long and short amounts from two simultaneous equations for target sensitivities (Numerical)
May 2026	Q4(a)	6	Sharpe single-index model optimal portfolio with given cut-off rate: securities selected, proportions, CAPM expected return and value, consistency with CAPM (Numerical)

What follows, exactly as ICAI printed it in the AFM study material: (1) Test Your Knowledge theory questions, with the SM paragraph ICAI's answer points to; (2) every SM illustration with ICAI's solution, grouped by SM section; (3) Test Your Knowledge practical questions with ICAI's solutions.

Ⓣ Theory questions (SM Test Your Knowledge) with ICAI answers

Theory Q1

1. Write short note on factors affecting decision of investment in fixed income securities.

ICAI answer: "Please refer paragraph 12.3" – SM paragraph 12.3 reproduced

12.3 Selection of Securities

There are certain criteria which must be kept in mind while selecting securities. The selection criteria for both bonds and equity shares are given as following:

12.3.1 Selection of Bonds

Bonds are fixed income avenues. The following factors have to be evaluated in selecting fixed income avenues:

(a) *Yield to maturity*: The yield to maturity for a fixed income avenues represent the rate of return earned by the investor, if he invests in the fixed income avenues and holds it till its maturity.

(b) *Risk of Default*: To assess such risk on a bond, one has to look at the credit rating of the bond. If no credit rating is available relevant financial ratios of the firm have to be examined such as debt equity, interest coverage, earning power etc and the general prospect of the industry to which the firm belongs have to be assessed.

(c) *Tax Shield:* In the past, several fixed income avenues offers tax shields but at present only a few of them do so.

(d) *Liquidity:* If the fixed income avenues can be converted wholly or substantially into cash at a fairly short notice it possesses a liquidity of a high order.

12.3.2 Selection of Stock (Equity Share)

Three approaches are applied for selection of equity shares- Technical analysis, Fundamental analysis and Random selection analysis.

(a) Technical analysis looks at price behaviours and volume data to determine whether the share price will move up or down or remain trend less.

(b) Fundamental analysis focuses on fundamental factors like earning level, growth prospects and risk exposure to establish intrinsic value of a share. The recommendation to buy hold or sell is based on comparison of intrinsic value and prevailing market price.

(c) Random selection analysis is based on the premise that the market is efficient and security is properly priced.

Levels of Market Efficiency and Approach to Security Selection

Approach / Levels of Efficiency	Technical Analysis	Fundaments Analysis	Random Selection
1) Inefficiency	Best	Poor	Poor
2) Weak form efficiency	Poor	Best	Poor
3) Semi-strong efficiency	Poor	Good	Fair
4) Strong Form efficiency	Poor	Fair	Best

Theory Q2

2. Briefly explain the objectives of "Portfolio Management".

ICAI answer: "Please refer paragraph 1.2" – SM paragraph 1.2 reproduced

1.2 Objectives of Portfolio Management

Some of the important objectives of portfolio management are:

(i) ***Security/Safety of Principal Amount*:** Security not only involves keeping the principal sum intact but also its purchasing power.

(ii) ***Stability of Income*:** To facilitate planning more accurately and systematically the reinvestment or consumption of income.

(iii) ***Capital Growth*:** It can be attained by reinvesting in growth securities or through purchase of growth securities.

(iv) ***Marketability i.e. the case with which a security can be bought or sold*:** This is essential for providing flexibility to investment portfolio.

(v) ***Liquidity i.e. nearness to money*:** It is desirable for the investor so as to take advantage of attractive opportunities upcoming in the market.

(vi) ***Diversification*:** The basic objective of building a portfolio is to reduce the risk of loss of capital and/or income by investing in various types of securities and over a wide range of industries.

(vii) ***Favourable Tax Status*:** The effective yield an investor gets from his investment depends on tax to which it is subjected to. By minimising the tax burden, yield can be effectively improved.

Theory Q3

3. Discuss the Capital Asset Pricing Model (CAPM) and its relevant assumptions.

ICAI answer: "Please refer paragraph 8" – SM paragraph 8 reproduced

8. CAPITAL ASSET PRICING MODEL (CAPM)

The CAPM distinguishes between risk of holding a single asset and holding a portfolio of assets. There is a trade-off between risk and return. Modern portfolio theory concentrates on risk and stresses on risk management rather than on return management. Risk may be security risk involving danger of loss of return from an investment in a single financial or capital asset. Security risk differs from portfolio risk, which is the probability of loss from investment in a portfolio of assets. Portfolio risk is comprised of unsystematic risk and systematic risk. Unsystematic risks can be averted through diversification and is related to random variables. Systematic risk is market related component of portfolio risk. It is commonly measured by regression coefficient Beta or the Beta coefficient. Low Beta reflects low risk and high Beta reflects high risk.

As the unsystematic risk can be diversified by building a portfolio, the relevant risk for determining the prices of securities is the non-diversifiable component of the total risk. As mentioned earlier, it can be measured by using Beta (β) a statistical parameter which measures the market sensitivity of returns. The beta for the market is equal to 1.0. Beta explains the systematic relationship between the return on a security and the return on the market by using a simple linear regression equation. The return on a security is taken as a dependent variable and the return on market is taken as independent variable then $R_j = R_f + \beta (R_m - R_f)$. The beta parameter (β) in this William Sharpe model represents the slope of the above regression relationship and measures the sensitivity or responsiveness of the security returns to the general market returns. The portfolio beta is merely the weighted average of the betas of individual securities included in the portfolio.

Portfolio Beta (β_P) = $\sum$ Proportion of Security × Beta for Security.

CAPM provides a conceptual framework for evaluating any investment decision where capital is committed with a goal of producing future returns. CAPM is based on certain assumptions to provide conceptual framework for evaluating risk and return. Some of the important assumptions are discussed below:

(i) ***Efficient market:*** It is the first assumption of CAPM. Efficient market refers to the existence of competitive market where financial securities and capital assets are bought and sold with full information of risk and return available to all participants. In an efficient market, the price of individual assets will reflect a real or intrinsic value of a share as the market prices will adjust quickly to any new situation, John J. Hampton has remarked in "Financial decision making" that although efficient capital market is not much relevant to capital budgeting decisions, but CAPM would be useful to evaluate capital budgeting proposal because the company can compare risk and return to be obtained by investment in machinery with risk and return from investment in securities.

(ii) ***Rational investment goals:*** Investors desire higher return for any acceptable level of risk or the lowest risk for any desired level of return. Such a rational choice is made on logical and consistent ranking of proposals in order of preference for higher good to lower good and this is the scale of the marginal efficiency of capital. Beside, transactive preferences and certainty equivalents are other parameters of rational choice.

(iii) Risk aversion in efficient market is adhered to although at times risk seeking behaviour is adopted for gains.

(iv) CAPM assumes that all assets are divisible and liquid assets.

(v) Investors are able to borrow freely at a risk less rate of interest i.e. borrowings can fetch equal return by investing in safe Government securities.

(vi) Securities can be exchanged without payment of brokerage, commissions or taxes and without any transaction cost.

(vii) Securities or capital assets face no bankruptcy or insolvency.

(viii) CAPM assumes that the Capital Market is in equilibrium.

Based on above assumptions the CAPM is developed with the main goal to formulate the return required by investors from a single investment or a portfolio of assets. The required rate of return is defined as the minimum expected return needed so that investors will purchase and hold an asset.

Risk and return relationship in this model stipulates higher return for higher level of risk and *vice versa*. However, there may be exception to this general rule where markets are not efficient.

Three aspects are worth consideration:

(a) Stock market is not concerned with diversifiable risk

(b) It is not concerned with an investor having a diversified portfolio

(c) Compensation paid is restricted to non-diversifiable risk.

Thus, an investor has to look into the non-diversifiable portion of risk on one side and returns on the other side. To establish a link between the two, the required return one expects to get for a given level of risk has been mandated by the Capital Asset Pricing Model.

If the risk free investment R_f is 5%, an investor can earn this return of 5% by investing in risk free investment. Again if the stock market earns a rate of return R_m which is 15% then an investor investing in stocks constituting the stock market index will earn also 15%. Thus, the excess return earned over and above the risk free return is called the market risk premium ($R_m - R_f$) i.e. (15% - 5%) = 10% which is the reward for undertaking risk. So, if an investment is as risky as the stock market, the risk premium to be earned is 10%.

If an investment is 30% riskier than the stock market, it would carry risk premium i.e. 30% more than the risk premium of the stock market i.e. 10% + 30% of 10% = 10% + 3% = 13%. β identifies how much more risky is an investment with reference to the stock market. Hence the risk premium that a stock should earn is β times the risk premium from the market [$\beta \times (R_m - R_f)$]. The total return from an investment is the risk free rate of return plus the risk premium. So the required return from a stock would be $R_j = R_f + [\beta \times (R_m - R_f)]$. In the above example $5\% + 1.3 \times (15\% - 5\%) = 18\%$

The risk premium on a stock varies in direct proportion to its Beta. If the market risk premium is 6% and β of a stock is 1.2 then the risk premium for that stock shall be 7.2% (6% × 1.2)

Illustration 2

A company's beta is 1.40. The market return is 14% and the risk free rate is 10%.

(i) What is the expected return of the company's stock based on CAPM.

(ii) If the risk premium on the market goes up by 2.50% points, what would be the revised expected return on this stock?

Solution

(i) Computation of expected return based on CAPM

$R_j = R_f + \beta (R_m - R_f) = 10\% + 1.40 (14\% - 10\%) = 10\% + 5.6\% = 15.6\%$

(ii) Computation of expected return if the market risk premium goes up by 2.50% points

The return from the market goes up by 2.50% i.e. 14% + 2.50% = 16.50%

Expected Return based on CAPM is given by

R_j = 10% + 1.40 (16.5% - 10%) = 10% + 1.40 × 6.5% = 10% + 9.10% = 19.10%

A graphical representation of CAPM is the Security Market Line, (SML). This line indicates the rate of return required to compensate at a given level of risk. Plotting required return on Y axis and Beta on the X-axis we get an upward sloping line which is given by (R_m – R_f), the risk premium.

The higher the Beta value of a security, higher would be the risk premium relative to the market. This upward sloping line is called the Security Market Line. It measures the relationship between systematic risk and return.

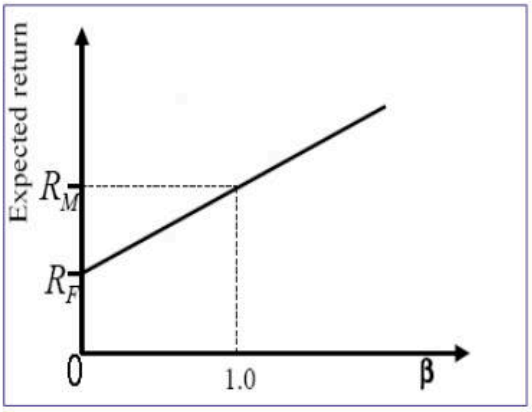


Illustration 3

The risk premium for the market is 10%. Assuming Beta values of Security K are 0, 0.25, 0.42, 1.00 and 1.67. Compute the risk premium on Security K.

Solution

Market Risk Premium is 10%

β Value of K	*Risk Premium of K*
0.00	0%
0.25	2.50%
0.42	4.20%
1.00	10.00%
1.67	16.70%

Illustration 4

Treasury Bills give a return of 5%. Market Return is 13%

(i) What is the market risk premium

(ii) Compute the β Value and required returns for the following combination of investments.

Treasury Bill	100	70	30	0
Market	0	30	70	100

Solution

Risk Premium $R_m - R_f$ = 13% - 5% = 8%

β is the weighted average investing in portfolio consisting of market β = 1 and treasury bills (β = 0)

Portfolio	*Treasury Bills: Market*	*β*	*$R_j = R_f + \beta \times (R_m - R_f)$*
1	100:0	0	5% + 0(13% - 5%) = 5%
2	70:30	0.7(0) + 0.3(1) = 0.3	5% + 0.3(13% - 5%) = 7.40%
3	30:70	0.3(0) + 0.7(1) = 0.7	5% + 0.7(13% - 5%) = 10.60%
4	0:100	1	5% + 1.0(13% - 5%) = 13%

8.1 Risk free Rate of Return

In CAPM, there is only one risk free rate. It presumes that the returns on a security include both dividend payments and capital appreciation. These require to be factored in judging the value of Beta and in computing the required rate of return.

Illustration 5

Pearl Ltd. expects that considering the current market prices, the equity shareholders as per Moderate Approach, should get a return of at least 15.50% while the current return on the market is 12%. RBI has closed the latest auction for ₹ 2500 crores of 182 day bills for the lowest bid of 4.3% although there were bidders at a higher rate of 4.6% also for lots of less than ₹ 10 crores. What is Pearl Ltd's Beta?

Solution

Determining Risk free rate: Two risk free rates are given. The aggressive approach would be to consider 4.6% while the conservative approach would be to take 4.3%. If we take the moderate value then the simple average of the two i.e. 4.45% would be considered

Application of CAPM

$$R_j = R_f + \beta (R_m - R_f)$$

$$15.50\% = 4.45\% + \beta (12\% - 4.45\%)$$

$$\beta = \frac{15.50\% - 4.45\%}{12\% - 4.45\%} = \frac{11.05}{7.55}$$

$$= 1.464$$

Illustration 6

The following information is available with respect of Jaykay Ltd.

Year	Jay Kay Limited		Market		Return on Govt. Bonds
	Average Share Price (₹)	DPS (₹)	Average Index	Dividend Yield (%)	
2002	242	20	1812	4	6
2003	279	25	1950	5	5
2004	305	30	2258	6	4
2005	322	35	2220	7	5

Compute Beta Value of the company as at the end of 2005. What is your observation?

Solution

(1) Computation of Beta Value

Calculation of Returns

$$\text{Returns} = \frac{D_1 + (P_1 - P_0)}{P_0} \times 100$$

Year	*Returns*
2002 – 2003	$\frac{25 + (279 - 242)}{242} \times 100 = 25.62\%$
2003 – 2004	$\frac{30 + (305 - 279)}{279} \times 100 = 20.07\%$
2004 – 2005	$\frac{35 + (322 - 305)}{305} \times 100 = 17.05\%$

Calculation of Returns from market Index

Year	*% of Index Appreciation*	*Dividend Yield %*	*Total Return %*
2002 – 2003	$\frac{1950-1812}{1812}\times 100 = 7.62\%$	5%	12.62%
2003 – 2004	$\frac{2258-1950}{1950}\times 100 = 15.79\%$	6%	21.79%
2004 – 2005	$\frac{2220-2258}{2258}\times 100 = (-)1.68\%$	7%	5.32%

Computation of Beta

Year	*X*	*Y*	*XY*	*Y^2*
2002-2003	25.62	12.62	323.32	159.26
2003-2004	20.07	21.79	437.33	474.80
2004-2005	17.05	5.32	90.71	28.30
	62.74	39.73	851.36	662.36

$$\bar{X} = \frac{62.74}{3} = 20.91,\ \bar{Y} = \frac{39.73}{3} = 13.24$$

$$\beta = \frac{\sum XY - n\,\bar{X}\,\bar{Y}}{\sum Y^2 - n(\bar{Y})^2}$$

$$= \frac{851.36 - 3(20.91)(13.24)}{662.36 - 3(13.24)^2}$$

$$= \frac{851.36 - 830.55}{662.36 - 525.89} = \frac{20.81}{136.47} = 0.15$$

(2) Observation: Based on the beta value of the company and the analysis of returns, it can be concluded that changes in the return on share of Jaykay Ltd. are less affected by changes in market returns.

8.2 Under Valued and Over Valued Stocks

The CAPM model can be practically used to buy, sell or hold stocks. CAPM provides the required rate of return on a stock after considering the risk involved in an investment. Based on current market price or any other judgmental factors (benchmark) one can identify as to what would be the expected return over a period of time. By comparing the required return as per CAPM with the expected return the following investment decisions are available:

(a) **When required return as per CAPM < Actual Return – Buy:** This is due to the stock being undervalued i.e. the stock gives more return than what it should give.

(b) **When required return as per CAPM > Actual Return – Sell:** This is due to the stock being overvalued i.e. the stock gives less return than what it should give.

(c) **When required return as per CAPM = Actual Return – Hold:** This is due to the stock being correctly valued i.e. the stock gives same return than what it should give.

From another angle, if the current market price is considered as a basis of CAPM then:

(i) Actual Market Price < Market Price using CAPM, stock is undervalued

(ii) Actual market Price > Market Price using CAPM, stock is overvalued

(iii) Actual market Price = Market Price using CAPM, stock is correctly valued.

Illustration 7

Information related to an investment is as follows:

Risk free rate	10%
Market Return	15%
Beta	1.2

(i) What would be the return from this investment?

(ii) If the projected return is 18%, is the investment rightly valued?

(iii) What is your strategy?

Solution

(i) Required rate of Return as per CAPM is given by

$R_j = R_f + \beta (R_m - R_f)$

$= 10 + 1.2 (15-10) = 16\%$

(ii) Since projected return is 18%, the stock is not rightly valued rather undervalued as return as per CAPM less than Projected Return.

(iii) Had this Project Return is considered as expected return, the decision should be to BUY the share.

Illustration 8

The expected returns and Beta of three stocks are given below

Stock	A	B	C
Expected Return (%)	18	11	15
Beta Factor	1.7	0.6	1.2

If the risk free rate is 9% and the expected rate of return on the market portfolio is 14% which of the above stocks are over, under or correctly valued in the market? What shall be the strategy?

Solution

Required Rate of Return is given by

$$R_j = R_f + \beta (R_m - R_f)$$

For Stock A, $R_j = 9 + 1.7 (14 - 9) = 17.50\%$

Stock B, $R_j = 9 + 0.6 (14-9) = 12.00\%$

Stock C, $R_j = 9 + 1.2 (14-9) = 15.00\%$

Required Return %	*Expected Return %*	*Valuation*	*Decision*
17.50%	18.00%	Under Valued	Buy
12.00%	11.00%	Over Valued	Sell
15.00%	15.00%	Correctly Valued	Hold

8.3 Advantages and Limitations of CAPM

The advantages of CAPM can be listed as:

(i) *Risk Adjusted Return:* It provides a reasonable basis for estimating the required return on an investment which has risk inbuilt into it. Hence, it can also be used as Risk Adjusted Discount Rate in Capital Budgeting.

(ii) *No Dividend Company*: It is useful in computing the cost of equity of a company which does not declare dividend.

There are certain limitations of CAPM as well, which are discussed as follows:

(a) *Reliability of Beta*: Statistically reliable Beta might not exist for shares of many firms. It may not be possible to determine the cost of equity of all firms using CAPM. All shortcomings that apply to Beta value applies to CAPM too.

(b) *Other Risks*: By emphasising on systematic risk only, the unsystematic risk that are of importance too does not find place in a diversified portfolio.

(c) *Information Available*: It is extremely difficult to obtain important information on risk free interest rate and expected return on market portfolio because depending upon the situations there can be multiple risk free rates and market being volatile its return varies from time to time.

Ⓘ SM illustrations with ICAI solutions

7. Single Index Model (Sharpe Index Model)

SM Ch 6 · Illustration 1

Illustration 1

The following details are given for X and Y companies' stocks and the Bombay Sensex for a period of one year. Calculate the systematic and unsystematic risk for the companies' stocks. If equal amount of money is allocated for the stocks what would be the portfolio risk?

	X Stock	Y Stock	Sensex
Average return	0.15	0.25	0.06
Variance of return	6.30	5.86	2.25
β	0.71	0.685	
Correlation Co-efficient	0.424		
Co-efficient of determination (r^2)	0.18		

Solution — ICAI

Solution

The co-efficient of determination (r^2) i.e. square of Coefficient of Correlation gives the percentage of the variation in the security's return that is explained by the variation of the market index return. In the X company stock return, 18 per cent of variation is explained by the variation of the index and 82 per cent is not explained by the index.

According to Sharpe, the variance explained by the index is the systematic risk. The unexplained variance or the residual variance is the unsystematic risk.

Company X:

Systematic risk = $\beta_i^2 \times$ Variance of market index

= $(0.71)^2 \times 2.25 = 1.134$

Unsystematic risk($\in_i^2$) = Total variance of security return - Systematic Risk

= 6.30 – 1.134

= 5.166

or

= Variance of Security Return $(1-r^2)$

= 6.30 X (1-0.18) = 6.3 X 0.82 = 5.166

Total risk = $\beta_i^2 \times \sigma_m^2 + \in_i^2$

= Systematic Risk + Unsystematic Risk

= 1.134 + 5.166 = 6.30

Company Y:

Systematic risk = $\beta_i^2 \times \sigma_m^2$

= $(0.685)^2$ x 2.25 = 1.056

Unsystematic risk = Total variance of the security return - systematic risk.

= 5.86-1.056 = 4.804

Portfolio Risk

$$\sigma_p^2 = \left[\left(\sum_{i=1}^{N} X_i \beta_i\right)^2 \sigma_m^2\right] + \left[\left(\sum_{i=1}^{N} X_i^2 \epsilon_i^2\right)\right]$$

= $[(0.5 \times 0.71 + 0.5 \times 0.685)^2\ 2.25] + [(0.5)^2(5.166)+(0.5)^2(4.804)]$

= $[(0.355 + 0.3425)^2\ 2.25] + [(1.292 + 1.201)]$

= 1.0946 + 2.493 = 3.5876

8. Capital Asset Pricing Model (Capm)

SM Ch 6 · Illustration 2

Illustration 2

A company's beta is 1.40. The market return is 14% and the risk free rate is 10%.

(i) What is the expected return of the company's stock based on CAPM.

(ii) If the risk premium on the market goes up by 2.50% points, what would be the revised expected return on this stock?

Solution — ICAI

Solution

(i) Computation of expected return based on CAPM

$R_j = R_f + \beta (R_m - R_f)$ = 10% + 1.40 (14% - 10%) = 10% + 5.6% = 15.6%

(ii) Computation of expected return if the market risk premium goes up by 2.50% points

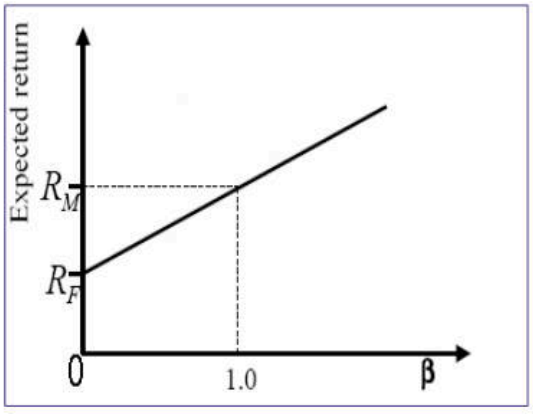
The return from the market goes up by 2.50% i.e. 14% + 2.50% = 16.50%

Expected Return based on CAPM is given by

R_j = 10% + 1.40 (16.5% - 10%) = 10% + 1.40 × 6.5% = 10% + 9.10% = 19.10%

A graphical representation of CAPM is the Security Market Line, (SML). This line indicates the rate of return required to compensate at a given level of risk. Plotting required return on Y axis and Beta on the X-axis we get an upward sloping line which is given by (R_m – R_f), the risk premium.

The higher the Beta value of a security, higher would be the risk premium relative to the market. This upward sloping line is called the Security Market Line. It measures the relationship between systematic risk and return.



SM Ch 6 · Illustration 3

Illustration 3

The risk premium for the market is 10%. Assuming Beta values of Security K are 0, 0.25, 0.42, 1.00 and 1.67. Compute the risk premium on Security K.

Solution — ICAI

Solution

Market Risk Premium is 10%

β Value of K	Risk Premium of K
0.00	0%
0.25	2.50%
0.42	4.20%
1.00	10.00%
1.67	16.70%

SM Ch 6 · Illustration 4

Illustration 4

Treasury Bills give a return of 5%. Market Return is 13%

(i) What is the market risk premium

(ii) Compute the β Value and required returns for the following combination of investments.

Treasury Bill	100	70	30	0
Market	0	30	70	100

Solution — ICAI

Solution

Risk Premium $R_m - R_f = 13\% - 5\% = 8\%$

β is the weighted average investing in portfolio consisting of market $\beta = 1$ and treasury bills ($\beta = 0$)

Portfolio	Treasury Bills: Market	β	$R_j = R_f + \beta \times (R_m - R_f)$
1	100:0	0	5% + 0(13% - 5%) = 5%
2	70:30	0.7(0) + 0.3(1) = 0.3	5% + 0.3(13% - 5%) = 7.40%
3	30:70	0.3(0) + 0.7(1) = 0.7	5% + 0.7(13% - 5%) = 10.60%
4	0:100	1	5% + 1.0(13% - 5%) = 13%

SM Ch 6 · Illustration 5 (under 8.1 Risk free Rate of Return)

Illustration 5

Pearl Ltd. expects that considering the current market prices, the equity shareholders as per Moderate Approach, should get a return of at least 15.50% while the current return on the market is 12%. RBI has closed the latest auction for ₹ 2500 crores of 182 day bills for the lowest bid of 4.3% although there were bidders at a higher rate of 4.6% also for lots of less than ₹ 10 crores. What is Pearl Ltd's Beta?

Solution — ICAI

Solution

Determining Risk free rate: Two risk free rates are given. The aggressive approach would be to consider 4.6% while the conservative approach would be to take 4.3%. If we take the moderate value then the simple average of the two i.e. 4.45% would be considered

Application of CAPM

$$R_j = R_f + \beta (R_m - R_f)$$

$$15.50\% = 4.45\% + \beta (12\% - 4.45\%)$$

$$\beta = \frac{15.50\% - 4.45\%}{12\% - 4.45\%} = \frac{11.05}{7.55}$$

$$= 1.464$$

SM Ch 6 · Illustration 6 (under 8.1 Risk free Rate of Return)

Illustration 6

The following information is available with respect of Jaykay Ltd.

Year	Jay Kay Limited		Market		Return on Govt. Bonds
	Average Share Price (₹)	DPS (₹)	Average Index	Dividend Yield (%)	
2002	242	20	1812	4	6
2003	279	25	1950	5	5
2004	305	30	2258	6	4
2005	322	35	2220	7	5

Compute Beta Value of the company as at the end of 2005. What is your observation?

Solution — ICAI

Solution

(1) Computation of Beta Value

Calculation of Returns

$$\text{Returns} = \frac{D_1 + (P_1 - P_0)}{P_0} \times 100$$

Year	Returns
2002 – 2003	$\frac{25 + (279 - 242)}{242} \times 100 = 25.62\%$
2003 – 2004	$\frac{30 + (305 - 279)}{279} \times 100 = 20.07\%$
2004 – 2005	$\frac{35 + (322 - 305)}{305} \times 100 = 17.05\%$

Calculation of Returns from market Index

Year	% of Index Appreciation	Dividend Yield %	Total Return %
2002 – 2003	$\frac{1950 - 1812}{1812} \times 100 = 7.62\%$	5%	12.62%
2003 – 2004	$\frac{2258 - 1950}{1950} \times 100 = 15.79\%$	6%	21.79%
2004 – 2005	$\frac{2220 - 2258}{2258} \times 100 = (-)1.68\%$	7%	5.32%

Computation of Beta

Year	X	Y	XY	Y^2
2002-2003	25.62	12.62	323.32	159.26
2003-2004	20.07	21.79	437.33	474.80
2004-2005	17.05	5.32	90.71	28.30
	62.74	39.73	851.36	662.36

$$\bar{X} = \frac{62.74}{3} = 20.91, \quad \bar{Y} = \frac{39.73}{3} = 13.24$$

$$\beta = \frac{\sum XY - n\,\bar{X}\,\bar{Y}}{\sum Y^2 - n(\bar{Y})^2}$$

$$= \frac{851.36 - 3(20.91)(13.24)}{662.36 - 3(13.24)^2}$$

$$= \frac{851.36 - 830.55}{662.36 - 525.89} = \frac{20.81}{136.47} = 0.15$$

(2) Observation: Based on the beta value of the company and the analysis of returns, it can be concluded that changes in the return on share of Jaykay Ltd. are less affected by changes in market returns.

SM Ch 6 · Illustration 7 (under 8.2 Under Valued and Over Valued Stocks)

Illustration 7

Information related to an investment is as follows:

Risk free rate	10%
Market Return	15%
Beta	1.2

(i) What would be the return from this investment?

(ii) If the projected return is 18%, is the investment rightly valued?

(iii) What is your strategy?

Solution — ICAI

Solution

(i) Required rate of Return as per CAPM is given by

$R_j = R_f + \beta (R_m - R_f)$

$= 10 + 1.2 (15-10) = 16\%$

(ii) Since projected return is 18%, the stock is not rightly valued rather undervalued as return as per CAPM less than Projected Return.

(iii) Had this Project Return is considered as expected return, the decision should be to BUY the share.

SM Ch 6 · Illustration 8 (under 8.2 Under Valued and Over Valued Stocks)

Illustration 8

The expected returns and Beta of three stocks are given below

Stock	A	B	C
Expected Return (%)	18	11	15
Beta Factor	1.7	0.6	1.2

If the risk free rate is 9% and the expected rate of return on the market portfolio is 14% which of the above stocks are over, under or correctly valued in the market? What shall be the strategy?

Solution — ICAI

Solution

Required Rate of Return is given by

$R_j = R_f + \beta (R_m - R_f)$

For Stock A, $R_j = 9 + 1.7 (14 - 9) = 17.50\%$

Stock B, $R_j = 9 + 0.6 (14-9) = 12.00\%$

Stock C, $R_j = 9 + 1.2 (14-9) = 15.00\%$

Required Return %	Expected Return %	Valuation	Decision
17.50%	18.00%	Under Valued	Buy
12.00%	11.00%	Over Valued	Sell
15.00%	15.00%	Correctly Valued	Hold

9. Arbitrage Pricing Theory Model (Apt)

SM Ch 6 · Illustration 9

Illustration 9

With the help of following data determine the return on the security X.

Factor	Risk Premium associated with the Factor	β_i
Market	4%	1.3
Growth Rate of GDP	1%	0.3
Inflation	-4%	0.2

Risk Free Rate of Return is 8%.

Solution — ICAI

Solution

Expected Return = $R_f + \lambda_1\beta_1 + \lambda_2\beta_2 + \lambda_3\beta_3$

= 8% + 1.3 x 4% + 0.3 x 1% + 0.2 x (-4%)

= 8% + 5.2% + 0.3% - 0.8%

= 12.7%

As mentioned earlier while CAPM concentrates on one factor (market risk) in its Model, APT does not specifically requires any particular type of factor to be concentrated upon. Though Stephan Ross identified change in the following factors:

- Inflation
- Level of Industrial Production
- Risk Premium
- Term Structure of Interest Rates

Further according to Ross, if no surprise happens to these macro-economic factors then actual returns shall be equal to expected. In case, if any unanticipated changes happens in these factors, then formula of APT shall be as follows:

$E(R) = R_f + \beta_1 (EV_1 - AV_1) + \beta_2 (EV_2 - AV_2) + \ldots\ldots\ldots \beta_n (EV_n - AV_n)$

Where

$(EV_n - AV_n)$ = Surprise Factor due to change in the Value of Factor

R_f = Risk Free Rate of Return

β_n = Sensitivity of corresponding Macro-economic factor

Ⓟ Practical questions (SM Test Your Knowledge) with ICAI solutions

Practical Q1

1. A stock costing ₹ 120 pays no dividends. The possible prices that the stock might sell for at the end of the year with the respective probabilities are:

Price	Probability
115	0.1
120	0.1
125	0.2
130	0.3
135	0.2
140	0.1

Required:

(i) Calculate the expected return.

(ii) Calculate the Standard deviation of returns.

Solution — ICAI

1. (i) Here, the probable returns have to be calculated using the formula

$$R = \frac{D}{P_0} + \frac{P_1 - P_0}{P_0}$$

Calculation of Probable Returns

Possible prices (P_1) ₹	P_1-P_0 ₹	[(P_1-P_0)/ P_0] x 100 Return (per cent)
115	-5	-4.17
120	0	0.00
125	5	4.17
130	10	8.33
135	15	12.50
140	20	16.67

Alternatively, it can be calculated as follows:

Calculation of Expected Returns

Possible return X_i	Probability $p(X_i)$	Product X_1-$p(X_i)$
-4.17	0.1	-0.417
0.00	0.1	0.000
4.17	0.2	0.834
8.33	0.3	2.499
12.50	0.2	2.500
16.67	0.1	1.667
		X = 7.083

Expected return X = 7.083 per

Alternatively, it can also be calculated as follows:

Expected Price = 115 x 0.1 + 120 x 0.1 + 125 x 0.2 + 130 x 0.3 + 135 x 0.2 + 140 x 0.1 = 128.50

$$\text{Return} = \frac{128.50 - 120}{120} \times 100 = 7.0833\%$$

(ii) **Calculation of Standard Deviation of Returns**

Probable return X_i	Probability $p(X_i)$	Deviation $(X_i - X)$	Deviation squared $(X_i - X)^2$	Product $(X_i - X)^2 p(X_i)$
-4.17	0.1	-11.253	126.63	12.66
0.00	0.1	-7.083	50.17	5.017
4.17	0.2	-2.913	8.49	1.698
8.33	0.3	1.247	1.56	0.467
12.50	0.2	5.417	29.34	5.869
16.67	0.1	9.587	91.91	9.191
				$\sigma^2 = 34.902$

Variance, σ^2 = 34.902 per cent

Standard deviation, $\sigma = \sqrt{34.902}$ = 5.908 per cent

Practical Q2

2. Following information is available in respect of expected dividend, market price and market condition after one year.

Market condition	Probability	Market Price	Dividend per share
		₹	₹
Good	0.25	115	9
Normal	0.50	107	5
Bad	0.25	97	3

The existing market price of equity share is ₹ 106 (F.V. ₹ 1), which is cum 10% bonus debenture of ₹ 6 each, per share. M/s. X Finance Company Ltd. had offered the buy-back of debentures at face value.

Find out the expected return and variability of returns of the equity shares if buyback offer is accepted by the investor.

And also advise-Whether to accept buy-back offer?

Solution — ICAI

2. **The Expected Return of the equity share may be found as follows:**

Market Condition	Probability	Total Return	Cost (*)	Net Return
Good	0.25	₹ 124	₹ 100	₹ 24
Normal	0.50	₹ 112	₹ 100	₹ 12
Bad	0.25	₹ 100	₹ 100	₹ 0

Expected Return = $(24 \times 0.25) + (12 \times 0.50) + (0 \times 0.25) = 12 = \left(\frac{12}{100}\right) \times 100 = 12\%$

The variability of return can be calculated in terms of standard deviation.

V SD $= 0.25\,(24 - 12)^2 + 0.50\,(12 - 12)^2 + 0.25\,(0 - 12)^2$

$= 0.25\,(12)^2 + 0.50\,(0)^2 + 0.25\,(-12)^2$

$= 36 + 0 + 36$

SD $= \sqrt{72}$

SD $= 8.485$ or say 8.49

(*) The present market price of the share is ₹ 106 cum bonus 10% debenture of ₹ 6 each; hence the net cost is ₹ 100.

M/s X Finance company has offered the buyback of debenture at face value. There is reasonable 10% rate of interest compared to expected return 12% from the market. Considering the dividend rate and market price the creditworthiness of the company seems to be very good. The decision regarding buy-back should be taken considering the maturity period and opportunity in the market. Normally, if the maturity period is low say up to 1 year better to wait otherwise to opt buy back option.

Practical Q3

3. Mr. A is interested to invest ₹ 1,00,000 in the securities market. He selected two securities B and D for this purpose. The risk return profile of these securities are as follows :

Security	Risk (σ)	Expected Return (ER)
B	10%	12%
D	18%	20%

Co-efficient of correlation between B and D is 0.15.

You are required to calculate the portfolio return of the following portfolios of B and D to be considered by A for his investment.

(i) 100 percent investment in B only;

(ii) 50 percent of the fund in B and the rest 50 percent in D;

(iii) 75 percent of the fund in B and the rest 25 percent in D; and

(iv) 100 percent investment in D only.

Also indicate that which portfolio is best for him from risk as well as return point of view?

Solution — ICAI

3. We have $E_p = W_1E_1 + W_3E_3 + \ldots\ldots\ldots W_nE_n$

and for standard deviation $\sigma^2_p = \sum_{i=1}^{n}\sum_{j=1}^{n} w_i w_j \sigma_{ij}$

$$\sigma^2_p = \sum_{i=1}^{n}\sum_{j=1}^{n} w_i w_j \rho_{ij}\, \sigma_i\, \sigma_j$$

Two asset portfolio

$\sigma^2_p = w^2_1\sigma^2_1 + w^2_2\sigma^2_2 + 2\, w_1w_2\sigma_1\sigma_2\rho_{12}$

Substituting the respective values we get,

(i) All funds invested in B

$Ep = 12\%$

$\sigma_p = 10\%$

(ii) 50% of funds in each of B & D

$Ep = 0.50X12\%+0.50X20\%=16\%$

$\sigma^2_p = (0.50)^2(10\%)^2 + (0.50)^2(18\%)^2 +2(0.50)(0.50)(0.15)(10\%)(18\%)$

$\sigma^2_p = 25 + 81 + 13.5 = 119.50$

$\sigma_p = 10.93\%$

(iii) 75% in B and 25% in D

$Ep = 0.75\%X12\%+0.25\%X20=14\%$

$\sigma^2_p = (0.75)^2(10\%)^2 + (0.25)^2(18\%)^2 +2(0.75)(0.25)(0.15)(10\%)(18\%)$

$\sigma^2_p = 56.25 + 20.25 + 10.125 = 86.625$

$\sigma_p = 9.31\%$

(iv) All funds in D

$Ep = 20\%$

$\sigma_p = 18.0\%$

Portfolio	(i)	(ii)	(iii)	(iv)
Return	12	16	14	20
σ	10	10.93	9.31	18

In the terms of return, we see that portfolio (iv) is the best portfolio. In terms of risk we see that portfolio (iii) is the best portfolio.

Practical Q4

4. Consider the following information on two stocks, A and B :

Year	Return on A (%)	Return on B (%)
2006	10	12
2007	16	18

You are required to determine:

(i) The expected return on a portfolio containing A and B in the proportion of 40% and 60% respectively.

(ii) The Standard Deviation of return from each of the two stocks.

(iii) The covariance of returns from the two stocks.

(iv) Correlation coefficient between the returns of the two stocks.

(v) The risk of a portfolio containing A and B in the proportion of 40% and 60%.

Solution — ICAI

4. (i) Expected return of the portfolio A and B

E (A) = (10 + 16) / 2 = 13%

E (B) = (12 + 18) / 2 = 15%

$$Rp = \sum_{i=1}^{N} X_i R_i = 0.4(13) + 0.6(15) = 14.2\%$$

(ii) Stock A:

Variance = 0.5 (10 – 13)² + 0.5 (16 – 13) ² = 9

Standard deviation = $\sqrt{9}$ = 3%

Stock B:

Variance = 0.5 (12 – 15) ² + 0.5 (18 – 15) ² = 9

Standard deviation = 3%

(iii) Covariance of stocks A and B

Cov_{AB} = 0.5 (10 – 13) (12 – 15) + 0.5 (16 – 13) (18 – 15) = 9

(iv) Correlation of coefficient

$$r_{AB} = \frac{Cov_{AB}}{\sigma_A \sigma_B} = \frac{9}{3 \times 3} = 1$$

(v) Portfolio Risk

$$\sigma_P = \sqrt{X^2{}_A \sigma^2{}_A + X^2{}_B \sigma^2{}_B + 2X_A X_B (\sigma_A \sigma_B \sigma_{AB})}$$

$$= \sqrt{(0.4)^2 (3)^2 + (0.6)^2 (3)^2 + 2(0.4)(0.6)(3)(3)(1)}$$

$$= \sqrt{1.44 + 3.24 + 4.32} = 3\%$$

Practical Q5

5. Following is the data regarding six securities:

	A	B	C	D	E	F
Return (%)	8	8	12	4	9	8
Risk (Standard deviation)	4	5	12	4	5	6

(i) Assuming three will have to be selected, state which ones will be picked.

(ii) Assuming perfect correlation, show whether it is preferable to invest 75% in A and 25% in C or to invest 100% in E.

Solution — ICAI

5. (i) Security A has a return of 8% for a risk of 4, whereas B and F have a higher risk for the same return. Hence, among them A dominates.

For the same degree of risk 4, security D has only a return of 4%. Hence, D is also dominated by A.

Securities C and E remain in reckoning as they have a higher return though with higher degree of risk.

Hence, the ones to be selected are A, C & E.

(ii) The average values for A and C for a proportion of 3 : 1 will be :

$$\text{Risk} = \frac{(3\times 4)+(1\times 12)}{4} = 6\%$$

$$\text{Return} = \frac{(3\times 8)+(1\times 12)}{4} = 9\%$$

Therefore:	75% A	E
	25% C	–
Risk	6	5
Return	9%	9%

For the same 9% return the risk is lower in E. Hence, E will be preferable.

Practical Q6

6. The historical rates of return of two securities over the past ten years are given. Calculate the Covariance and the Correlation coefficient of the two securities:

Years:	1	2	3	4	5	6	7	8	9	10
Security 1: (Return per cent)	12	8	7	14	16	15	18	20	16	22
Security 2: (Return per cent)	20	22	24	18	15	20	24	25	22	20

Solution — ICAI

6. **Calculation of Covariance**

Year	R_1	Deviation $(R_1-\overline{R_1})$	Deviation $(R_1-\overline{R_1})^2$	R_2	Deviation $(R_2-\overline{R_2})$	Deviation $(R_2-\overline{R_2})^2$	Product of deviations
1	12	-2.8	7.84	20	-1	1	2.8
2	8	-6.8	46.24	22	1	1	-6.8
3	7	-7.8	60.84	24	3	9	-23.4
4	14	-0.8	0.64	18	-3	9	2.4
5	16	1.2	1.44	15	-6	36	-7.2
6	15	0.2	0.04	20	-1	1	-0.2
7	18	3.2	10.24	24	3	9	9.6
8	20	5.2	27.04	25	4	16	20.8
9	16	1.2	1.44	22	1	1	1.2
10	22	7.2	51.84	20	-1	1	-7.2
$\overline{R_1}=\frac{148}{10}=14.8$			Σ=207.60		$\overline{R_2}=\frac{210}{10}=21$	Σ=84.00	

$$\text{Covariance} = \frac{\sum_{i=1}^{N}[R_1-\overline{R_1}][R_2-\overline{R_2}]}{N} = -8/10 = -0.8$$

Standard Deviation of Security 1

$$\sigma_1=\sqrt{\frac{(R_1-\overline{R_1})^2}{N}}$$

$$\sigma_1=\sqrt{\frac{207.60}{10}}=\sqrt{20.76}$$

$$\sigma_1 = 4.56$$

Standard Deviation of Security 2

$$\sigma_2=\sqrt{\frac{(R_2-\overline{R_2})^2}{N}}$$

$$\sigma_2=\sqrt{\frac{84}{10}}=\sqrt{8.40}$$

$$\sigma_2 = 2.90$$

Alternatively, Standard Deviation of securities can also be calculated as follows:

Calculation of Standard Deviation

Year	R_1	R_1^2	R_2	R_2^2
1	12	144	20	400
2	8	64	22	484
3	7	49	24	576
4	14	196	18	324
5	16	256	15	225
6	15	225	20	400

7	18	324	24	576
8	20	400	25	625
9	16	256	22	484
10	22	484	20	400
	148	2398	210	4494

Standard deviation of security 1:

$$\sigma_1 = \sqrt{\frac{N\sum R_1^2 - (\sum R_1)^2}{N^2}}$$

$$= \sqrt{\frac{(10\times 2398) - (148)^2}{10\times 10}} = \sqrt{\frac{23980 - 21904}{100}}$$

$$= \sqrt{20.76} = 4.56$$

Standard deviation of security 2:

$$\sigma_2 = \sqrt{\frac{N\sum R_2^2 - (\sum R_2)^2}{N^2}}$$

$$= \sqrt{\frac{(10\times 4494) - (210)^2}{10\times 10}} = \sqrt{\frac{44940 - 44100}{100}}$$

$$= \sqrt{\frac{840}{100}} = \sqrt{8.4} = 2.90$$

Correlation Coefficient

$$r_{12} = \frac{Cov}{\sigma_1 \sigma_2} = \frac{-0.8}{4.56\times 2.90} = \frac{-0.8}{13.22} = -0.0605$$

Practical Q7

7. An investor has decided to invest to invest ₹ 1,00,000 in the shares of two companies, namely, ABC and XYZ. The projections of returns from the shares of the two companies along with their probabilities are as follows:

Probability	ABC(%)	XYZ(%)
0.20	12	16
0.25	14	10
0.25	-7	28
0.30	28	-2

You are required to

(i) Comment on return and risk of investment in individual shares.

(ii) Compare the risk and return of these two shares with a Portfolio of these shares in equal proportions.

(iii) Find out the proportion of each of the above shares to formulate a minimum risk portfolio.

Solution — ICAI

7. (i)

Probability	ABC (%)	XYZ (%)	1X2 (%)	1X3 (%)
(1)	(2)	(3)	(4)	(5)
0.20	12	16	2.40	3.2
0.25	14	10	3.50	2.5
0.25	-7	28	-1.75	7.0
0.30	28	-2	8.40	-0.6
	Average return		12.55	12.1

Hence the expected return from ABC = 12.55% and XYZ is 12.1%

Probability	(ABC- $\overline{ABC}$)	(ABC-$\overline{ABC}$)^2	1X3	(XYZ-$\overline{XYZ}$)	(XYZ-$\overline{XYZ}$)^2	(1)X(6)
(1)	(2)	(3)	(4)	(5)	(6)	
0.20	-0.55	0.3025	0.06	3.9	15.21	3.04
0.25	1.45	2.1025	0.53	-2.1	4.41	1.10
0.25	-19.55	382.2025	95.55	15.9	252.81	63.20
0.30	15.45	238.7025	71.61	-14.1	198.81	59.64
			167.75			126.98

σ^2_{ABC} = 167.75(%)^2 ; σ_{ABC} = 12.95%

σ^2_{XYZ} = 126.98(%)^2 ; σ_{XYZ} = 11.27%

(ii) In order to find risk of portfolio of two shares, the covariance between the two is necessary here.

Probability	(ABC-$\overline{ABC}$)	(XYZ-$\overline{XYZ}$)	2X3	1X4
(1)	(2)	(3)	(4)	(5)
0.20	-0.55	3.9	-2.145	-0.429
0.25	1.45	-2.1	-3.045	-0.761
0.25	-19.55	15.9	-310.845	-77.71
0.30	15.45	-14.1	-217.845	-65.35
				-144.25

σ^2_P = (0.5^2 x 167.75) + (0.5^2 x 126.98) + 2 x (-144.25) x 0.5 x 0.5

σ^2_P = 41.9375 + 31.745 – 72.125

σ^2_P = 1.5575 or 1.56(%)

$\sigma_P = \sqrt{1.56}$ = 1.25%

E (R_p) = (0.5 x 12.55) + (0.5 x 12.1) = 12.325%

Hence, the return is 12.325% with the risk of 1.25% for the portfolio. Thus, the portfolio results in the reduction of risk by the combination of two shares.

(iii) For constructing the minimum risk portfolio the condition to be satisfied is

$$X_{ABC} = \frac{\sigma_X^2 - r_{AX}\sigma_A\sigma_X}{\sigma_A^2 + \sigma_X^2 - 2r_{AX}\sigma_A\sigma_X} \text{ or } = \frac{\sigma_X^2 - Cov._{AX}}{\sigma_A^2 + \sigma_X^2 - 2Cov._{AX}}$$

σ_X = Std. Deviation of XYZ

σ_A = Std. Deviation of ABC

r_{AX}= Coefficient of Correlation between XYZ and ABC

$Cov._{AX}$ = Covariance between XYZ and ABC.

Therefore,

$$\% \text{ ABC} = \frac{126.98-(-144.25)}{126.98+167.75-[2\times(-144.25)]} = \frac{271.23}{583.23} = 0.46 \text{ or } 46\%$$

% ABC = 46%, XYZ = 54%

(1 – 0.46) =0.54

Practical Q8

8. The following information are available with respect of Krishna Ltd.

Year	Krishna Ltd. Average share price	Dividend per Share	Average Market Index	Dividend Yield	Return on Govt. bonds
	₹	₹			
2012	245	20	2013	4%	7%
2013	253	22	2130	5%	6%
2014	310	25	2350	6%	6%
2015	330	30	2580	7%	6%

Compute Beta Value of the Krishna Ltd. at the end of 2015 and state your observation.

Solution — ICAI

8. (i) Computation of Beta Value

Calculation of Returns

$$\text{Returns} = \frac{D_1 + (P_1 - P_0)}{P_0} \times 100$$

Year	Returns
2012 – 13	$\frac{22 + (253 - 245)}{245} \times 100 = 12.24\%$
2013 – 14	$\frac{25 + (310 - 253)}{253} \times 100 = 32.41\%$
2014 – 15	$\frac{30 + (330 - 310)}{310} \times 100 = 16.13\%$

Calculation of Returns from market Index

Year	% of Index Appreciation	Dividend Yield %	Total Return %
2012–13	$\frac{(2130 - 2013)}{2013} \times 100 = 5.81\%$	5%	10.81%

2013–14	$\frac{(2350-2130)}{2130}\times 100 = 10.33\%$	6%	16.33%
2014–15	$\frac{(2580-2350)}{2350}\times 100 = 9.79\%$	7%	16.79%

Computation of Beta

Year	Krishna Ltd. (X)	Market Index (Y)	XY	Y^2
2012–13	12.24%	10.81%	132.31	116.86
2013–14	32.41%	16.33%	529.25	266.67
2014–15	16.13%	16.79%	270.82	281.90
Total	60.78%	43.93%	932.38	665.43

Average Return of Krishna Ltd. = $\frac{60.78}{3} = 20.26\%$

Average Market Return = $\frac{43.93}{3} = 14.64\%$

Beta (β) = $\frac{\sum XY - n\bar{X}\bar{Y}}{\sum Y^2 - n(\bar{Y})^2} = \frac{932.38 - 3\times 20.26\times 14.64}{665.43 - 3(14.64)^2} = 1.897$

(ii) Observation

	Expected Return (%)	Actual Return (%)	Action
2012 – 13	6%+ 1.897(10.81% - 6%) = 15.12%	12.24%	Sell
2013 – 14	6%+ 1.897(16.33% - 6%) = 25.60%	32.41%	Buy
2014 – 15	6%+ 1.897(16.79% - 6%) = 26.47%	16.13%	Sell

Practical Q9

9. The distribution of return of security 'F' and the market portfolio 'P' is given below:

Probability	Return %	
	F	P
0.30	30	-10
0.40	20	20
0.30	0	30

You are required to calculate the expected return of security 'F' and the market portfolio 'P', the covariance between the market portfolio and security and beta for the security.

Solution — ICAI

9. Security F

Prob(P)	R_f	PxR_f	Deviations of F ($R_f - ER_f$)	(Deviation)² of F	(Deviations)² P_x
0.3	30	9	13	169	50.7
0.4	20	8	3	9	3.6
0.3	0	0	-17	289	86.7
		ER_f=17			Var_f =141

STDEV $\sigma_f = \sqrt{141} = 11.87$

Market Portfolio, P

R_M %	P_M	Exp. Return $R_M \times P_M$	Dev. of P (R_M-ER_M)	(Dev. of P)²	(DeV.)² P_M	(Deviation of F) x (Deviation of P)	Dev. of F x Dev. of P) x P
-10	0.3	-3	-24	576	172.8	-312	-93.6
20	0.4	8	6	36	14.4	18	7.2
30	0.3	9	16	256	76.8	-272	-81.6
		ER_M=14			Var $_M$=264 σ_M=16.25		=Co Var P_M =- 168

Beta= $\frac{\text{Co Var } P_M}{\sigma_M^2} = \frac{-168}{264} = -.636$

Practical Q10

10. Given below is information of market rates of Returns and Data from two Companies A and B:

	Year 2007	Year 2008	Year 2009
Market (%)	12.0	11.0	9.0
Company A (%)	13.0	11.5	9.8
Company B (%)	11.0	10.5	9.5

You are required to determine the beta coefficients of the Shares of Company A and Company B.

Solution — ICAI

10. Company A:

Year	*Return % (Ra)*	*Market return % (Rm)*	*Deviation R(a)*	*Deviation Rm*	*D Ra × DRm*	*Rm²*
1	13.0	12.0	1.57	1.33	2.09	1.77
2	11.5	11.0	0.07	0.33	0.02	0.11
3	9.8	9.0	–1.63	–1.67	2.72	2.79
	34.3	32.0			4.83	4.67

Average Ra = 11.43

Average Rm = 10.67

$$\text{Covariance} = \frac{\sum (R_m - \bar{R}_m)(R_a - \bar{R}_a)}{N}$$

$$\text{Covariance} = \frac{4.83}{3} = 1.61$$

$$\text{Variance } (\sigma_m^2) = \frac{\sum (R_m - \bar{R}_m)^2}{N}$$

$$= \frac{4.67}{3} = 1.557$$

$$\beta = \frac{1.61}{1.557} = 1.03$$

Company B:

Year	Return % (Rb)	Market return % (Rm)	Deviation R(b)	Deviation Rm	D Rb × D Rm	Rm²
1	11.0	12.0	0.67	1.33	0.89	1.77
2	10.5	11.0	0.17	0.33	0.06	0.11
3	9.5	9.0	–0.83	–1.67	1.39	2.79
	31.0	32.0			2.34	4.67

Average Rb = 10.33

Average Rm = 10.67

$$\text{Covariance} = \frac{\sum (R_m - \bar{R}_m)(R_b - \bar{R}_b)}{N}$$

$$\text{Covariance} = \frac{2.34}{3} = 0.78$$

$$\text{Variance } (\sigma_m^2) = \frac{\sum (R_m - \bar{R}_m)^2}{N}$$

$$= \frac{4.67}{3} = 1.557$$

$$\beta = \frac{0.78}{1.557} = 0.50$$

Practical QII

11. The returns on stock A and market portfolio for a period of 6 years are as follows:

Year	Return on A (%)	Return on market portfolio (%)
1	12	8
2	15	12
3	11	11
4	2	-4
5	10	9.5
6	-12	-2

You are required to determine:

(i) Characteristic line for stock A

(ii) The systematic and unsystematic risk of stock A.

Solution — ICAI

11. Characteristic line is given by

$\alpha + \beta$ Rm

$$\beta i = \frac{\Sigma xy - n\bar{x}\bar{y}}{\Sigma x^2 - n(\bar{x})^2}$$

$$\alpha i = \bar{y} - \beta \bar{x}$$

Return on A (Y)	Return on market (X)	xy	x^2	$(x-\bar{x})$	$(x-\bar{x})^2$	$(y-\bar{y})$	$(y-\bar{y})^2$
12	8	96	64	2.25	5.06	5.67	32.15
15	12	180	144	6.25	39.06	8.67	75.17
11	11	121	121	5.25	27.56	4.67	21.81
2	-4	-8	16	-9.75	95.06	-4.33	18.75
10	9.5	95	90.25	3.75	14.06	3.67	13.47
-12	-2	24	4	-7.75	60.06	-18.33	335.99
38	34.5	508	439.25		240.86		497.34

$\bar{y} = 38/6 = 6.33$

$\bar{x} = 34.5/6 = 5.75$

$$\beta = \frac{\Sigma xy - n\bar{x}\bar{y}}{\Sigma x^2 - n(\bar{x})^2} = \frac{508 - 6(5.75)(6.33)}{439.25 - 6(5.75)^2} = \frac{508 - 218.385}{439.25 - 198.375}$$

$$= \frac{289.615}{240.875} = 1.202$$

$\alpha = \bar{y} - \beta \bar{x} = 6.33 - 1.202\ (5.75) = -0.58$

Hence the characteristic line is -0.58 + 1.202 (R_m)

Total Risk of Market = $\sigma_{m^2} = \frac{\Sigma(x-\bar{x})^2}{n} = \frac{240.86}{6} = 40.14(\%)$

Total Risk of Stock= $\frac{497.34}{6} = 82.89\ (\%)$

Systematic Risk = $\beta i^2\ \sigma_2 = (1.202)^2 \times 40.14 = 57.99(\%)$

Unsystematic Risk is = Total Risk – Systematic Risk

= 82.89 - 57.99 = 24.90(%)

Practical Q12

12. The rates of return on the security of Company X and market portfolio for 10 periods are given below:

Period	Return of Security X (%)	Return on Market Portfolio (%)
1	20	22
2	22	20
3	25	18
4	21	16
5	18	20
6	–5	8
7	17	–6
8	19	5
9	–7	6
10	20	11

(i) What is the beta of Security X?

(ii) What is the characteristic line for Security X?

Solution — ICAI

12. (i)

Period	R_X	R_M	$R_X - \bar{R}_X$	$R_M - \bar{R}_M$	$(R_X - \bar{R}_X)(R_M - \bar{R}_M)$	$(R_M - \bar{R}_M)^2$
1	20	22	5	10	50	100
2	22	20	7	8	56	64
3	25	18	10	6	60	36
4	21	16	6	4	24	16
5	18	20	3	8	24	64
6	-5	8	-20	-4	80	16
7	17	-6	2	-18	-36	324

8	19	5	4	-7	-28	49
9	-7	6	-22	-6	132	36
10	20	11	5	-1	-5	1
	150	120			357	706
	ΣR_X	ΣR_M			$\sum (R_X - \bar{R}_X)(R_M - \bar{R}_M)$	$\sum (R_M - \bar{R}_M)^2$

$\bar{R}_X = 15$ $\bar{R}_M = 12$

$$\sigma^2_M = \frac{\sum\left(R_M - \bar{R}_M\right)^2}{n} = \frac{706}{10} = 70.60$$

$$Cov_{XM} = \frac{\sum\left(R_X - \bar{R}_X\right)\left(R_M - \bar{R}_M\right)}{n} = \frac{357}{10} = 35.70$$

$$Beta_x = \frac{Cov_{XM}}{\sigma^2_M} = \frac{35.70}{70.60} = 0.505$$

Alternative Solution

Period	X	Y	Y^2	XY
1	20	22	484	440
2	22	20	400	440
3	25	18	324	450
4	21	16	256	336
5	18	20	400	360
6	-5	8	64	-40
7	17	-6	36	-102
8	19	5	25	95
9	-7	6	36	-42
10	20	11	121	220
	150	120	2146	2157
	$\bar{X} = 15$	$\bar{Y} = 12$		

$$= \frac{\Sigma XY - n\bar{X}\bar{Y}}{\Sigma X^2 - n(\bar{X})^2}$$

$$= \frac{2157 - 10 \times 15 \times 12}{2146 - 10 \times 12 \times 12} = \frac{357}{706} = 0.506$$

(ii) $\bar{R}_X = 15$ $\bar{R}_M = 12$

$y = \alpha + \beta x$

$15 = \alpha + 0.505 \times 12$

Alpha (α) $= 15 - (0.505 \times 12) = 8.94\%$

Characteristic line for security X = $\alpha + \beta \times R_M$

Where, R_M = Expected return on Market Index

$\therefore$ Characteristic line for security X = $8.94 + 0.505\ R_M$

Practical Q13

13. Expected returns on two stocks for particular market returns are given in the following table:

Market Return	Aggressive	Defensive
7%	4%	9%
25%	40%	18%

You are required to calculate:

(a) The Betas of the two stocks.

(b) Expected return of each stock, if the market return is equally likely to be 7% or 25%.

(c) The Security Market Line (SML), if the risk free rate is 7.5% and market return is equally likely to be 7% or 25%.

(d) The Alphas of the two stocks.

Solution — ICAI

13. (a) The Betas of two stocks:

Aggressive stock - (40% - 4%)/(25% - 7%) = 2

Defensive stock - (18% - 9%)/(25% - 7%) = 0.50

Alternatively, it can also be solved by using the Characteristic Line Relationship as follows:

$R_s = \alpha + \beta R_m$

Where

α = Alpha

β = Beta

R_m= Market Return

For Aggressive Stock

$4\% = \alpha + \beta(7\%)$

$40\% = \alpha + \beta(25\%)$

$36\% = \beta(18\%)$

$\beta = 2$

For Defensive Stock

$9\% = \alpha + \beta(7\%)$

$18\% = \alpha + \beta(25\%)$

$9\% = \beta(18\%)$

$\beta = 0.50$

(b) Expected returns of the two stocks:-

Aggressive stock - 0.5 x 4% + 0.5 x 40% = 22%

Defensive stock - 0.5 x 9% + 0.5 x 18% = 13.5%

(c) Expected return of market portfolio = 0.5 x 7% + 0.5% x 25% = 16%

$\therefore$ Market risk prem. = 16% - 7.5% = 8.5%

$\therefore$ SML is, required return = 7.5% + βi 8.5%

(d) $R_s = \alpha + \beta R_m$

For Aggressive Stock

$22\% = \alpha_A + 2(16\%)$

$\alpha_A = -10\%$

For Defensive Stock

$13.5\% = \alpha_D + 0.50(16\%)$

$\alpha_D = 5.5\%$

Practical Q14

14. A study by a Mutual fund has revealed the following data in respect of three securities:

Security	σ (%)	Correlation with Index, Pm
A	20	0.60
B	18	0.95
C	12	0.75

The standard deviation of market portfolio (BSE Sensex) is observed to be 15%.

(i) What is the sensitivity of returns of each stock with respect to the market?

(ii) What are the covariances among the various stocks?

(iii) What would be the risk of portfolio consisting of all the three stocks equally?

(iv) What is the beta of the portfolio consisting of equal investment in each stock?

(v) What is the total, systematic and unsystematic risk of the portfolio in (iv)?

Solution — ICAI

14. (i) Sensitivity of each stock with market is given by its beta.

Standard deviation of market Index = 15%

Variance of market Index = 0.0225

Beta of stocks = $\sigma_i r / \sigma_m$

A = 20 × 0.60/15 = 0.80

B = 18 × 0.95/15 = 1.14

C = 12 × 0.75/15 = 0.60

(ii) Covariance between any 2 stocks = $\beta_1 \beta_2 \sigma^2_m$

Covariance matrix

Stock/Beta	0.80	1.14	0.60
A	400.000	205.200	108.000
B	205.200	324.000	153.900
C	108.000	153.900	144.000

(iii) Total risk of the equally weighted portfolio (Variance) = $400(1/3)^2 + 324(1/3)^2 + 144(1/3)^2 + 2(205.20)(1/3)^2 + 2(108.0)(1/3)^2 + 2(153.900)(1/3)^2 = 200.244$

(iv) β of equally weighted portfolio $= \beta_p = \sum \beta_i / N = \frac{0.80 + 1.14 + 0.60}{3}$

$= 0.8467$

(v) Systematic Risk $\beta_p^2 \sigma_m^2 = (0.8467)^2 (15)^2 = 161.302$

Unsystematic Risk = Total Risk – Systematic Risk

= 200.244 – 161.302 = 38.942

Practical QIS

15. Mr. X owns a portfolio with the following characteristics:

	Security A	Security B	Risk Free security
Factor 1 sensitivity	0.80	1.50	0
Factor 2 sensitivity	0.60	1.20	0
Expected Return	15%	20%	10%

It is assumed that security returns are generated by a two factor model.

(i) If Mr. X has ₹ 1,00,000 to invest and sells short ₹ 50,000 of security B and purchases ₹ 1,50,000 of security A what is the sensitivity of Mr. X's portfolio to the two factors?

(ii) If Mr. X borrows ₹ 1,00,000 at the risk free rate and invests the amount he borrows along with the original amount of ₹ 1,00,000 in security A and B in the same proportion as described in part (i), what is the sensitivity of the portfolio to the two factors?

(iii) What is the expected return premium of factor 2?

Solution — ICAI

15. (i) Mr. X's position in the two securities are +1.50 in security A and -0.5 in security B. Hence the portfolio sensitivities to the two factors:-

b prop. 1 =1.50 x 0.80 + (-0.50 x 1.50) = 0.45

b prop. 2 = 1.50 x 0.60 + (-0.50 x 1.20) = 0.30

(ii) Mr. X's revised position:-

Security A ₹ 3,00,000 / ₹ 1,00,000 = 3

Security B -₹ 1,00,000 / ₹ 1,00,000 = -1

Risk free asset -₹ 100000 / ₹ 100000 = -1

b prop. 1 = 3.0 x 0.80 + (-1 x 1.50) + (- 1 x 0) = 0.90

b prop. 2 = 3.0 x 0.60 + (-1 x 1.20) + (-1 x 0) = 0.60

(iii) Expected Return = Risk Free Rate of Return + Risk Premium

Let λ_1 and λ_2 are the Value Factor 1 and Factor 2 respectively.

Accordingly

$15 = 10 + 0.80\ \lambda_1 + 0.60\ \lambda_2$

$20 = 10 + 1.50\ \lambda_1 + 1.20\ \lambda_2$

On solving equation, the value of $\lambda_1 = 0$, and risk premium of factor 2 for Securities A & B shall be as follows:

Using Security A's Return

Total Return = 15% = 10% + 0.60 λ_2

Risk Premium (λ_2) = 5%/ 0.60 = 8.33%

Alternatively using Security B's Return

Total Return = 20% = 10 + 1.20 λ_2

Risk Premium = 10%/ 1.20 = 8.33%

Practical Q16

16. Mr. Tempest has the following portfolio of four shares:

Name	Beta	Investment ₹ Lac.
Oxy Rin Ltd.	0.45	0.80
Boxed Ltd.	0.35	1.50
Square Ltd.	1.15	2.25
Ellipse Ltd.	1.85	4.50

The risk-free rate of return is 7% and the market rate of return is 14%.

Required.

(i) Determine the portfolio return. (ii) Calculate the portfolio Beta.

Solution — ICAI

16. Market Risk Premium (A) = 14% – 7% = 7%

Share	*Beta*	*Risk Premium (Beta x A) %*	*Risk Free Return %*	*Return %*	*Return ₹*
Oxy Rin Ltd.	0.45	3.15	7	10.15	8,120
Boxed Ltd.	0.35	2.45	7	9.45	14,175
Square Ltd.	1.15	8.05	7	15.05	33,863
Ellipse Ltd.	1.85	12.95	7	19.95	89,775
Total Return					1,45,933

Total Investment ₹ 9,05,000

(i) Portfolio Return = $\frac{₹\,1,45,933}{₹\,9,05,000} \times 100 = 16.13\%$

(ii) Portfolio Beta

Portfolio Return = Risk Free Rate + Risk Premium x β = 16.13%

7% + 7β = 16.13%

β = 1.30

Alternative Approach

First we shall compute Portfolio Beta using the weighted average method as follows:

$$Beta_P = 0.45 \times \frac{0.80}{9.05} + 0.35 \times \frac{1.50}{9.05} + 1.15 \times \frac{2.25}{9.05} + 1.85 \times \frac{4.50}{9.05}$$

$= 0.45 \times 0.0884 + 0.35 \times 0.1657 + 1.15 \times 0.2486 + 1.85 \times 0.4972$

$= 0.0398 + 0.058 + 0.2859 + 0.9198 = 1.3035$

Accordingly,

(i) Portfolio Return using CAPM formula will be as follows:

$R_P = R_F + \beta_P(R_M - R_F)$

$= 7\% + 1.3035(14\% - 7\%) = 7\% + 1.3035(7\%)$

$= 7\% + 9.1245\% = 16.1245\%$

(ii) Portfolio Beta

As calculated above 1.3035

Practical Q17

17. Mr. Abhishek is interested in investing ₹ 2,00,000 for which he is considering following three alternatives:

(i) Invest ₹ 2,00,000 in Mutual Fund X (MFX)

(ii) Invest ₹ 2,00,000 in Mutual Fund Y (MFY)

(iii) Invest ₹ 1,20,000 in Mutual Fund X (MFX) and ₹ 80,000 in Mutual Fund Y (MFY)

Average annual return earned by MFX and MFY is 15% and 14% respectively. Risk free rate of return is 10% and market rate of return is 12%.

Covariance of returns of MFX, MFY and market portfolio Mix are as follow:

	MFX	MFY	Mix
MFX	4.800	4.300	3.370
MFY	4.300	4.250	2.800
Mix	3.370	2.800	3.100

You are required to calculate:

(i) variance of return from MFX, MFY and market return,

(ii) portfolio return, beta, portfolio variance and portfolio standard deviation,

(iii) expected return, systematic risk and unsystematic risk; and

(iv) Sharpe ratio, Treynor ratio and Alpha of MFX, MFY and Portfolio Mix.

Solution — ICAI

17. (i) Variance of Returns

$$Cor_{i,j} = \frac{Cov(i,j)}{\sigma_i \sigma_j}$$

Accordingly, for MFX

$$1 = \frac{Cov(X,X)}{\sigma_X \sigma_X}$$

$$\sigma_X^2 = 4.800$$

Accordingly, for MFY

$$1 = \frac{Cov(Y,Y)}{\sigma_Y \sigma_Y}$$

$$\sigma_Y^2 = 4.250$$

Accordingly, for Market Return

$$1 = \frac{Cov(M,M)}{\sigma_M \sigma_M}$$

$$\sigma_M^2 = 3.100$$

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Alternatively, by referring diagonally the given Table these values can identified as follows:

$Variance_X$ = 4.800

$Variance_Y$ = 4.250

$Variance_M$ = 3.100

(ii) Portfolio return, beta, variance and standard deviation

Weight of MFX in portfolio = $\frac{1,20,000}{2,00,000}$ =0.60

Weight of MFY in portfolio = $\frac{80,000}{2,00,000}$ =0.40

Accordingly Portfolio Return

0.60 × 15% + 0.40 × 14% = 14.60%

Beta of each Fund

$$\beta = \frac{Cov(Fund, Market)}{Variance\ of\ Market}$$

$$\beta_X = \frac{3.370}{3.100} = 1.087$$

$$\beta_Y = \frac{2.800}{3.100} = 0.903$$

Portfolio Beta

0.60 x 1.087 + 0.40 x 0.903 = 1.013

Portfolio Variance

$\sigma^2_{XY} = w^2_X\sigma^2_X + w^2_Y\sigma^2_Y + 2\,w_X w_Y Cov_{X,Y}$

$= (0.60)^2 (4.800) + (0.40)^2 (4.250) + 2(0.60)\,(0.40)\,(4.300)$

= 4.472

Or Portfolio Standard Deviation

$\sigma_{XY} = \sqrt{4.472} = 2.115$

(iii) Expected Return, Systematic and Unsystematic Risk of Portfolio

Portfolio Return = 10% + 1.0134(12% - 10%) = 12.03%

MF X Return = 10% + 1.087(12% - 10%) = 12.17%

MF Y Return = 10% + 0.903(12% - 10%) = 11.81%

Systematic Risk = $\beta^2\sigma^2$

Accordingly,

Systematic Risk of MFX = $(1.087)^2$ x 3.10 = 3.663

Systematic Risk of MFY = $(0.903)^2$ x 3.10 = 2.528

Systematic Risk of Portfolio = $(1.013)^2$ x 3.10 = 3.181

Unsystematic Risk = Total Risk – Systematic Risk

Accordingly,

Unsystematic Risk of MFX = 4.80 – 3.663 = 1.137

Unsystematic Risk of MFY = 4.250 – 2.528 = 1.722

Unsystematic Risk of Portfolio = 4.472 – 3.181 = 1.291

(iv) Sharpe and Treynor Ratios and Alpha

Sharpe Ratio

$$\text{MFX} = \frac{15\% - 10\%}{\sqrt{4.800}} = 2.282$$

$$\text{MFY} = \frac{14\% - 10\%}{\sqrt{4.250}} = 1.94$$

$$\text{Portfolio} = \frac{14.6\% - 10\%}{2.115} = 2.175$$

Treynor Ratio

$$\text{MFX} = \frac{15\% - 10\%}{1.087} = 4.60$$

$$\text{MFY} = \frac{14\% - 10\%}{0.903} = 4.43$$

$$\text{Portfolio} = \frac{14.6\% - 10\%}{1.0134} = 4.54$$

Alpha

MFX = 15% - 12.17% = 2.83%

MFY = 14% - 11.81% = 2.19%

Portfolio = 14.6% - 12.03% = 2.57%

Practical Q18

18. Amal Ltd. has been maintaining a growth rate of 12% in dividends. The company has paid dividend @ ₹ 3 per share. The rate of return on market portfolio is 15% and the risk-free rate of return in the market has been observed as10%. The beta co-efficient of the company's share is 1.2.

You are required to calculate the expected rate of return on the company's shares as per CAPM model and the equilibrium price per share by dividend growth model.

Solution — ICAI

18. Capital Asset Pricing Model (CAPM) formula for calculation of expected rate of return is

$E_R = R_f + \beta (R_m - R_f)$

E_R = Expected Return

β = Beta of Security

R_m = Market Return

R_f = Risk free Rate

$= 10 + [1.2 (15 - 10)]$

$= 10 + 1.2 (5)$

$= 10 + 6 = 16\%$ or 0.16

Applying dividend growth mode for the calculation of per share equilibrium price:-

$$E_R = \frac{D_1}{P_0} + g$$

or $0.16 = \frac{3(1.12)}{P_0} + 0.12$ or $0.16 - 0.12 = \frac{3.36}{P_0}$

or $0.04\ P_0 = 3.36$ or $P_0 = \frac{3.36}{0.04} = ₹\ 84$

Therefore, equilibrium price per share will be ₹ 84.

Practical Q19

19. The following information is available in respect of Security X

Equilibrium Return	15%
Market Return	15%
7% Treasury Bond Trading at	$140
Covariance of Market Return and Security Return	225%
Coefficient of Correlation	0.75

You are required to determine the Standard Deviation of Market Return and Security Return.

Solution — ICAI

19. First we shall compute the β of Security X.

$$\text{Risk Free Rate} = \frac{\text{Coupon Payment}}{\text{Current Market Price}} = \frac{7}{140} = 5\%$$

Assuming equilibrium return to be equal to CAPM return then:

$15\% = R_f + \beta_X(R_m - R_f)$

$15\% = 5\% + \beta_X(15\% - 5\%)$

$\beta_X = 1$

or it can also be computed as follows:

$$\frac{R_m}{R_s} = \frac{15\%}{15\%} = 1$$

(i) Standard Deviation of Market Return

$$\beta_m = \frac{Cov_{X,m}}{\sigma_m^2} = \frac{225\%}{\sigma_m^2} = 1$$

$$\sigma^2_m = 225$$

$$\sigma_m = \sqrt{225} = 15\%$$

(ii) Standard Deviation of Security Return

$$\beta_X = \frac{\sigma_X}{\sigma_m} \times \rho_{Xm} = \frac{\sigma_X}{15} \times 0.75 = 1$$

$$\sigma_X = \frac{15}{0.75} = 20\%$$

Practical Q20

20. Assuming that shares of ABC Ltd. and XYZ Ltd. are correctly priced according to Capital Asset Pricing Model. The expected return from and Beta of these shares are as follows:

Share	Beta	Expected return
ABC	1.2	19.8%
XYZ	0.9	17.1%

You are required to derive Security Market Line.

Solution — ICAI

20. CAPM = $R_f + \beta (R_m - R_f)$

Accordingly

$R_{ABC} = R_f + 1.2 (R_m - R_f) = 19.8$

$R_{XYZ} = R_f + 0.9 (R_m - R_f) = 17.1$

$19.8 = R_f + 1.2 (R_m - R_f)$ ------(1)

$17.1 = R_f + 0.9 (R_m - R_f)$ ------(2)

Deduct (2) from (1)

$2.7 = 0.3 (R_m - R_f)$

$R_m - R_f = 9$

$R_f = R_m - 9$

Substituting in equation (1)

$19.8 = (R_m - 9) + 1.2 (R_m - R_m + 9)$

$19.8 = R_m - 9 + 10.8$

$19.8 = R_m + 1.8$

Then $R_m = 18\%$ and $R_f = 9\%$

Security Market Line

$= R_f + \beta$ (Market Risk Premium)

$= 9\% + \beta \times 9\%$

Practical Q21

21. A Ltd. has an expected return of 22% and Standard deviation of 40%. B Ltd. has an expected return of 24% and Standard deviation of 38%. A Ltd. has a beta of 0.86 and B Ltd. has a beta of 1.24. The correlation coefficient between the return of A Ltd. and B Ltd. is 0.72. The Standard deviation of the market return is 20%. Suggest:

(i) Is investing in B Ltd. better than investing in A Ltd.?

(ii) If you invest 30% in B Ltd. and 70% in A Ltd., what is your expected rate of return and portfolio Standard deviation?

(iii) What is the market portfolios expected rate of return and how much is the risk-free rate?

(iv) What is the beta of Portfolio if A Ltd.'s weight is 70% and B Ltd.'s weight is 30%?

Solution — ICAI

21. (i) A Ltd. has lower return and higher risk than B Ltd. Hence, investing in B Ltd. is better than in A Ltd. because the return is higher and the risk is lower. However, investing in both will yield diversification advantage.

(ii) $r_{AB} = 0.22 \times 0.7 + 0.24 \times 0.3 = 0.226$ i.e. 22.6%

$\sigma^2_{AB} = 0.40^2 \text{ X } 0.7^2 + 0.38^2 \text{ X } 0.3^2 + 2\text{X } 0.7 \text{ X } 0.3 \text{ X } 0.72 \text{ X } 0.40 \text{ X } 0.38 = 0.1374$

$\sigma_{AB} = \sqrt{\sigma^2_{AB}} = \sqrt{0.1374} = 0.37 = 37\%$

* Answer = 37.06% is also correct and variation may occur due to approximation.

(iii) This risk-free rate will be the same for A and B Ltd. Their rates of return are given as follows:

$r_A = 22 = r_f + (r_m - r_f)\ 0.86$

$r_B = 24 = r_f + (r_m - r_f)\ 1.24$

$r_A - r_B = -2 = (r_m - r_f)\ (-0.38)$

$r_m - r_f = -2/-0.38 = 5.26\%$

$r_A = 22 = r_f + (5.26)\ 0.86$

$r_f = 17.48\%$

Or

$r_B = 24 = r_f + (5.26)\ 1.24$

$r_f = 17.48\%$

$r_m - 17.48 = 5.26$

$r_m = 22.74\%$

(iv) $\beta_{AB} = \beta_A \times W_A + \beta_B \times W_B$

$= 0.86 \times 0.7 + 1.24 \times 0.3 = 0.974$

Practical Q22

22. XYZ Ltd. has substantial cash flow and until the surplus funds are utilised to meet the future capital expenditure, likely to happen after several months, are invested in a portfolio of short-term equity investments, details for which are given below:

Investment	No. of shares	Beta	Market price per share ₹	Expected dividend yield
I	60,000	1.16	4.29	19.50%
II	80,000	2.28	2.92	24.00%
III	1,00,000	0.90	2.17	17.50%
IV	1,25,000	1.50	3.14	26.00%

The current market return is 19% and the risk free rate is 11%.

Required to:

(i) Calculate the risk of XYZ's short-term investment portfolio relative to that of the market;

(ii) Whether XYZ should change the composition of its portfolio.

Solution — ICAI

22. (i) Computation of Beta of Portfolio

Investment	*No. of shares*	*Market Price*	*Market Value*	*Dividend Yield*	*Dividend*	*Composition*	*β*	*Weighted β*
I	60,000	4.29	2,57,400	19.50%	50,193	0.2339	1.16	0.27
II	80,000	2.92	2,33,600	24.00%	56,064	0.2123	2.28	0.48
III	1,00,000	2.17	2,17,000	17.50%	37,975	0.1972	0.90	0.18
IV	1,25,000	3.14	3,92,500	26.00%	1,02,050	0.3566	1.50	0.53
			11,00,500		2,46,282	1.0000		1.46

Return of the Portfolio $\frac{2,46,282}{11,00,500} = 0.2238$

Beta of Port Folio 1.46

Market Risk implicit

$0.2238 = 0.11 + \beta \times (0.19 - 0.11)$

Or, $0.08\ \beta + 0.11 = 0.2238$

$\beta = \frac{0.2238 - 0.11}{0.08} = 1.42$

Market β implicit is 1.42 while the port folio β is 1.46. Thus the portfolio is marginally risky compared to the market.

(ii) The decision regarding change of composition may be taken by comparing the dividend yield (given) and the expected return as per CAPM as follows:

Expected return R_s as per CAPM is:

$R_s = I_{RF} + (R_M - I_{RF})\beta$

For investment I $R_s = I_{RF} + (R_M - I_{RF})\beta$

$= .11 + (.19 - .11)\ 1.16$

$= 20.28\%$

For investment II, $R_s = .11 + (.19 - .11)\ 2.28 = 29.24\%$

For investment III, $R_s = .11 + (.19 - .11)\ .90$

$= 18.20\%$

For investment IV, $R_s = .11 + (.19 - .11)\ 1.50$

$= 23\%$

Comparison of dividend yield with the expected return R_s shows that the dividend yields of investment I, II and III are less than the corresponding R_s,. So, these investments are over-priced and should be sold by the investor. However, in case of investment IV, the dividend yield is more than the corresponding R_s, so, XYZ Ltd. should increase its proportion.

Practical Q23

23. A company has a choice of investments between several different equity oriented mutual funds. The company has an amount of ₹1 crore to invest. The details of the mutual funds are as follows:

Mutual Fund	Beta
A	1.6
B	1.0
C	0.9
D	2.0
E	0.6

Required:

(i) If the company invests 20% of its investment in each of the first two mutual funds and an equal amount in the mutual funds C, D and E, what is the beta of the portfolio?

(ii) If the company invests 15% of its investment in C, 15% in A, 10% in E and the balance in equal amount in the other two mutual funds, what is the beta of the portfolio?

(iii) If the expected return of market portfolio is 12% at a beta factor of 1.0, what will be the portfolios expected return in both the situations given above?

Solution — ICAI

23. With 20% investment in each MF Portfolio Beta is the weighted average of the Betas of various securities calculated as below:

(i)

Investment	*Beta (β)*	*Investment (₹ Lacs)*	*Weighted Investment*
A	1.6	20	32
B	1.0	20	20
C	0.9	20	18
D	2.0	20	40
E	0.6	20	12
		100	122
	Weighted *Beta (β)* = 1.22		

(ii) With varied percentages of investments portfolio beta is calculated as follows:

Investment	*Beta (β)*	*Investment (₹ Lacs)*	*Weighted Investment*
A	1.6	15	24
B	1.0	30	30
C	0.9	15	13.5
D	2.0	30	60
E	0.6	10	6
		100	133.5
	Weighted *Beta (β)* = 1.335		

(iii) Expected return of the portfolio with pattern of investment as in case (i)

= 12% × 1.22 i.e. 14.64%

Expected Return with pattern of investment as in case (ii) = 12% × 1.335 i.e., 16.02%.

Practical Q24

24. Suppose that economy A is growing rapidly and you are managing a global equity fund and so far you have invested only in developed-country stocks only. Now you have decided to add stocks of economy A to your portfolio. The table below shows the expected rates of return, standard deviations, and correlation coefficients (all estimates are for aggregate stock market of developed countries and stock market of Economy A).

	Developed Country Stocks	Stocks of Economy A
Expected rate of return (annualized percentage)	10	15
Risk [Annualized Standard Deviation (%)]	16	30
Correlation Coefficient (ρ)	0.30	

Assuming the risk-free interest rate to be 3%, you are required to determine:

(a) What percentage of your portfolio should you allocate to stocks of Economy A if you want to increase the expected rate of return on your portfolio by 0.5%?

(b) What will be the standard deviation of your portfolio assuming that stocks of Economy A are included in the portfolio as calculated above?

(c) Also show how well the Fund will be compensated for the risk undertaken due to inclusion of stocks of Economy A in the portfolio comparing with investment in developed country's stocks only.

Solution — ICAI

24. (a) Let the weight of stocks of Economy A be expressed as w, then

$(1- w)\times 10.0 + w \times 15.0 = 10.5$

i.e. $w = 0.1$ or 10%.

(b) Variance of portfolio shall be:

$(0.9)^2 (0.16)^2 + (0.1)^2 (0.30)^2 + 2(0.9)(0.1)(0.16)(0.30)(0.30) = 0.02423$

Standard deviation is $(0.02423)^{½} = 0.15565$ or 15.6%.

(c) The Sharpe ratio will improve by approximately 0.04, as shown below:

$$\text{Sharpe Ratio} = \frac{\text{Expected Return - Risk Free Rate of Return}}{\text{Standard Deviation}}$$

Investment only in developed countries: $\frac{10-3}{16} = 0.437$

With inclusion of stocks of Economy A: $\frac{10.5-3}{15.6} = 0.481$

Practical Q25

25. Mr. FedUp wants to invest an amount of ₹ 520 lakhs and had approached his Portfolio Manager. The Portfolio Manager had advised Mr. FedUp to invest in the following manner:

Security	Moderate	Better	Good	Very Good	Best
Amount (in ₹ Lakhs)	60	80	100	120	160
Beta	0.5	1.00	0.80	1.20	1.50

You are required to advise Mr. FedUp in regard to the following, using Capital Asset Pricing Methodology:

(i) Expected return on the portfolio, if the Government Securities are at 8% and the NIFTY is yielding 10%.

(ii) Advisability of replacing Security 'Better' with NIFTY.

Solution — ICAI

25. (i) Computation of Expected Return from Portfolio

Security	Beta (β)	Expected Return (r) as per CAPM	Amount (₹ Lakhs)	Weights (w)	wr
Moderate	0.50	8%+0.50(10% - 8%) = 9%	60	0.115	1.035
Better	1.00	8%+1.00(10% - 8%) = 10%	80	0.154	1.540
Good	0.80	8%+0.80 (10% - 8%) = 9.60%	100	0.192	1.843
Very Good	1.20	8%+1.20(10% - 8%) = 10.40%	120	0.231	2.402
Best	1.50	8%+1.50(10% - 8%) = 11%	160	0.308	3.388
Total			520	1	10.208

Thus Expected Return from Portfolio 10.208% say 10.21%.

Alternatively, it can be computed as follows:

$$\text{Average } \beta = 0.50 \times \frac{60}{520} + 1.00 \times \frac{80}{520} + 0.80 \times \frac{100}{520} + 1.20 \times \frac{120}{520} + 1.50 \times \frac{160}{520} = 1.104$$

As per CAPM

$= 0.08 + 1.104(0.10 - 0.08) = 0.10208$ i.e. 10.208%

(ii) As computed above the expected return from Better is 10% same as return from Nifty, hence there will be no difference even if the replacement of security is made. The main logic behind this neutrality is that the beta of security 'Better' is 1 which clearly indicates that this security shall yield same return as market return.

Practical Q26

26. Your client is holding the following securities:

Particulars of Securities	Cost	Dividends/Interest	Market price	Beta
	₹	₹	₹	
Equity Shares:				
Gold Ltd.	10,000	1,725	9,800	0.6
Silver Ltd.	15,000	1,000	16,200	0.8
Bronze Ltd.	14,000	700	20,000	0.6
GOI Bonds	36,000	3,600	34,500	0.01

Average return of the portfolio is 15.7%, calculate:

(i) Expected rate of return in each, using the Capital Asset Pricing Model (CAPM).

(ii) Risk free rate of return.

Solution — ICAI

26.

Particulars of Securities	*Cost ₹*	*Dividend*	*Capital gain*
Gold Ltd.	10,000	1,725	–200
Silver Ltd.	15,000	1,000	1,200
Bronz Ltd.	14,000	700	6,000

GOI Bonds	36,000	3,600	–1,500
Total	75,000	7,025	5,500

Expected rate of return on market portfolio

$$\frac{\text{Dividend Earned + Capital appreciation}}{\text{Initial investment}} \times 100$$

$$= \frac{₹\ 7,025 + ₹\ 5,500}{₹\ 75,000} \times 100 = 16.7\%$$

Risk free return

$$\text{Average of Betas} = \frac{0.6 + 0.8 + 0.6 + 0.01}{4} = \text{Average of Betas*} = 0.50$$

Average return = Risk free return + Average Betas (Expected return – Risk free return)

15.7 = Risk free return + 0.50 (16.7 – Risk free return)

Risk free return = 14.7%

* Alternatively, it can also be calculated through Weighted Average Beta.

Expected Rate of Return for each security is

Rate of Return = Rf + B (Rm – Rf)

Gold Ltd. = 14.7 + 0.6 (16.7 – 14.7)= 15.90%

Silver Ltd. = 14.7 + 0.8 (16.7 – 14.7) = 16.30%

Bronz Ltd. = 14.7 + 0.6 (16.7 – 14.7) = 15.90%

GOI Bonds = 14.7 + 0.01 (16.7 – 14.7) = 14.72%

* Alternatively, it can also be computed as follows:

Particulars of Securities (1)	Cost ₹ (2)	Market price ₹ (3)	Dividend ₹ (4)	Capital Gain ₹ (5)	Beta (6)	(3) x (6)
Gold Ltd.	10,000	9,800	1,725	–200	0.6	5,880
Silver Ltd.	15,000	16,200	1,000	1,200	0.8	12,960
Bronz Ltd.	14,000	20,000	700	6,000	0.6	12,000
GOI Bonds	36,000	34,500	3,600	–1,500	0.01	345
Total	75,000	80,500	7,025	5,500		31,185

$$\beta \text{ of the Portfolio} = \frac{31,185}{80,500} = 0.387$$

Using GOI Bond we can compute Risk Free Rate of Return as follows:

$$R_f = \frac{(34,500 - 36,000) + 3,600}{36,000} = 0.0583 \text{ i.e. } 5.83\%$$

Now we can calculate Market Return (R_m) using average return of the portfolio as follows:

$R_p = R_f + \beta(R_m - R_f)$

$15.7\% = 5.83\% + 0.387(R_m - 5.83\%)$

Practical Q27

27. A holds the following portfolio:

Share/Bond	Beta	Initial Price	Dividends	Market Price at end of year
		₹	₹	₹
Epsilon Ltd.	0.8	25	2	50
Sigma Ltd.	0.7	35	2	60
Omega Ltd.	0.5	45	2	135
GOI Bonds	0.01	1,000	140	1,005

Calculate:

(i) The expected rate of return of each security using Capital Asset Pricing Method (CAPM)

(ii) The average return of his portfolio.

Risk-free return is 14%.

Solution — ICAI

27. (i) Expected rate of return

	Total Investments	*Dividends*	*Capital Gains*
Epsilon Ltd.	25	2	25
Sigma Ltd.	35	2	25
Omega Ltd.	45	2	90
GOI Bonds	1,000	140	5
	1,105	146	145

Expected Return on market portfolio= $\frac{146+145}{1105}=26.33\%$

CAPM $E(Rp) = RF + \beta [E(RM) - RF]$

Epsilon Ltd	14+0.8 [26.33-14] =	14+9.86	=	23.86%
Sigma Ltd.	14+0.7 [26.33-14] =	14+8.63	=	22.63%
Omega Ltd.	14+0.5 [26.33-14] =	14+6.17	=	20.17%
GOI Bonds	14+0.01 [26.33-14] =	14+0.12	=	14.12%

(ii) Average Return of Portfolio

$$\frac{23.86+22.63+20.17+14.12}{4}=\frac{80.78}{4}=20.20\%$$

Alternatively, $\frac{0.8+0.7+0.5+0.01}{4}=\frac{2.01}{4}=0.5025$

14 + 0.5025 (26.33 - 14) = 14+ 6.20 = 20.20%

Practical Q28

28. Your client is holding the following securities:

Particulars of Securities	Cost ₹	Dividends ₹	Market Price ₹	BETA
Equity Shares:				
Co. X	8,000	800	8,200	0.8
Co. Y	10,000	800	10,500	0.7
Co. Z	16,000	800	22,000	0.5
PSU Bonds	34,000	3,400	32,300	0.2

Assuming a Risk-free rate of 15%, calculate:

- Expected rate of return in each, using the Capital Asset Pricing Model (CAPM).
- Simple Average return of the portfolio.

Solution — ICAI

28. Calculation of expected return on market portfolio (R_m)

Investment	Cost (₹)	Dividends (₹)	Capital Gains (₹)
Shares X	8,000	800	200
Shares Y	10,000	800	500
Shares Z	16,000	800	6,000
PSU Bonds	34,000	3,400	–1,700
	68,000	5,800	5,000

$$R_m = \frac{5,800 + 5,000}{68,000} \times 100 = 15.88\%$$

Calculation of expected rate of return on individual security:

Security		
Shares X	15 + 0.8 (15.88 – 15.0)	= 15.70%
Shares Y	15 + 0.7 (15.88 – 15.0)	= 15.62%
Shares Z	15 + 0.5 (15.88 – 15.0)	= 15.44%
PSU Bonds	15 + 0.2 (15.88 – 15.0)	= 15.18%

Calculation of the Average Return of the Portfolio:

$$= \frac{15.70 + 15.62 + 15.44 + 15.18}{4} = 15.49\%.$$

Practical Q29

29. An investor is holding 1,000 shares of Fatlass Company. Presently the rate of dividend being paid by the company is ₹ 2 per share and the share is being sold at ₹ 25 per share in the market. However, several factors are likely to be changed during the course of the year as indicated below:

	Existing	Revised
Risk free rate	12%	10%
Market risk premium	6%	4%
Beta value	1.4	1.25
Expected growth rate	5%	9%

In view of the above factors whether the investor should buy, hold or sell the shares? And why?

Solution — ICAI

29. On the basis of existing and revised factors, rate of return and price of share is to be calculated.

Existing rate of return

$= R_f + \text{Beta}\ (R_m - R_f) = 12\% + 1.4\ (6\%) = 20.4\%$

Revised rate of return

$= 10\% + 1.25\ (4\%) = 15\%$

Price of share (original)

$$P_o = \frac{D\ (1\ +\ g)}{K_e\ -\ g} = \frac{2\ (1.05)}{0.204\ -\ 0.05} = \frac{2.10}{0.154} = ₹\ 13.63$$

Price of share (Revised)

$$P_0 = \frac{2\ (1.09)}{0.15\ -\ 0.09} = \frac{2.18}{0.06} = ₹\ 36.33$$

In case of existing market price of ₹ 25 per share, rate of return (20.4%) and possible equilibrium price of share at ₹ 13.63, this share needs to be sold because the share is overpriced (₹ 25 – 13.63) by ₹ 11.37. However, under the changed scenario where growth of dividend has been revised at 9% and the return though decreased at 15% but the possible price of share is to be at ₹ 36.33 and therefore, in order to expect price appreciation to ₹ 36.33 the investor should hold the shares, if other things remain the same

Practical Q30

30. An investor is holding 5,000 shares of X Ltd. Current year dividend is ₹ 3 share. Market price of the share is ₹ 40 each. The investor is concerned about several factors which are likely to change during the next financial year as indicated below:

	Current Year	Next Year
Dividend paid /anticipated per share (₹)	3	2.5
Risk free rate	12%	10%
Market Risk Premium	5%	4%
Beta Value	1.3	1.4
Expected growth	9%	7%

In view of the above, advise whether the investor should buy, hold or sell the shares.

Solution — ICAI

30. On the basis of existing and revised factors, rate of return and price of share is to be calculated.

Existing rate of return

$= R_f + \text{Beta} (R_m - R_f)$

$= 12\% + 1.3 (5\%) = 18.5\%$

Revised rate of return

$= 10\% + 1.4 (4\%) = 15.60\%$

Price of share (original)

$$P_o = \frac{D (1 + g)}{K_e - g} = \frac{3 (1.09)}{0.185 - 0.09} = \frac{3.27}{0.095} = ₹ 34.42$$

Price of share (Revised)

$$P_o = \frac{2.50 (1.07)}{0.156 - 0.07} = \frac{2.675}{0.086} = ₹ 31.10$$

Market price of share of ₹ 40 is higher in comparison to current equilibrium price of ₹ 34.42 and revised equity price of ₹ 31.10. Under this situation investor should sell the share.

Practical Q31

31. An investor has two portfolios known to be on minimum variance set for a population of three securities A, B and C having below mentioned weights:

	WA	WB	WC
Portfolio X	0.30	0.40	0.30
Portfolio Y	0.20	0.50	0.30

It is supposed that there are no restrictions on short sales.

(i) What would be the weight for each stock for a portfolio constructed by investing ₹ 5,000 in portfolio X and ₹ 3,000 in portfolio Y?.

(ii) Suppose the investor invests ₹ 4,000 out of ₹ 8,000 in security A. How he will allocate the balance between security B and C to ensure that his portfolio is on minimum variance set?

Solution — ICAI

31. **(i)** Investment committed to each security would be:-

	A (₹)	B (₹)	C (₹)	Total (₹)
Portfolio X	1,500	2,000	1,500	5,000
Portfolio Y	600	1,500	900	3,000
Combined Portfolio	2,100	3,500	2,400	8,000
∴ Stock weights	0.26	0.44	0.30	

(ii) The equation of critical line takes the following form:-

WB = a + bWA

Substituting the values of WA & WB from portfolio X and Y in above equation, we get

0.40 = a + 0.30b, and

0.50 = a + 0.20b

Solving above equation we obtain the slope and intercept, a = 0.70 and b= -1 and thus, the critical line is

WB = 0.70 – WA

If half of the funds is invested in security A then,

WB = 0.70 – 0.50 = 0.20

Since WA + WB + WC = 1

WC = 1 - 0.50 – 0.20 = 0.30

∴ Allocation of funds to security B = 0.20 x 8,000 = ₹ 1,600, and

Security C = 0.30 x 8,000 = ₹ 2,400

Practical Q32

32. X Co., Ltd., invested on 1.4.2009 in certain equity shares as below:

Name of Co.	No. of shares	Cost (₹)
M Ltd.	1,000 (₹ 100 each)	2,00,000
N Ltd.	500 (₹ 10 each)	1,50,000

In September, 2009, 10% dividend was paid out by M Ltd. and in October, 2009, 30% dividend paid out by N Ltd. On 31.3.2010 market quotations showed a value of ₹ 220 and ₹ 290 per share for M Ltd. and N Ltd. respectively.

On 1.4.2010, investment advisors indicate (a) that the dividends from M Ltd. and N Ltd. for the year ending 31.3.2011 are likely to be 20% and 35%, respectively and (b) that the probabilities of market quotations on 31.3.2011 are as below:

Probability factor	Price/share of M Ltd.	Price/share of N Ltd.
0.2	220	290
0.5	250	310
0.3	280	330

You are required to:

(i) Calculate the average return from the portfolio for the year ended 31.3.2010;

(ii) Calculate the expected average return from the portfolio for the year 2010-11; and

(iii) Advise X Co. Ltd., of the comparative risk in the two investments by calculating the standard deviation in each case.

Solution — ICAI

32. Workings:

Calculation of return on portfolio for 2009-10	(Calculation in ₹ / share)		
	M	N	
Dividend received during the year	10	3	
Capital gain/loss by 31.03.10			
Market value by 31.03.10	220	290	
Cost of investment	200	300	
Gain/loss	20	(-)10	
Yield	30	(-)7	
Cost	200	300	
% return	15%	(-)2.33%	
Weight in the portfolio	57	43	
Weighted average return			7.55%
Calculation of estimated return for 2010-11			

Expected dividend	20	3.5	
Capital gain by 31.03.11			
(220x0.2)+ (250x0.5)+(280x0.3) – 220=(253-220)	33	-	
(290x0.2)+(310x0.5)+(330x0.3) – 290= (312 – 290)	-	22	
Yield	53	25.5	
*Market Value 01.04.10	220	290	
% return	24.09%	8.79%	
*Weight in portfolio (1,000x220): (500x290)	60.3	39.7	
Weighted average (Expected) return			18.02%
(*The market value on 31.03.10 is used as the base for calculating yield for 10-11)			

(i) Average Return from Portfolio for the year ended 31.03.2010 is 7.55%.

(ii) Expected Average Return from portfolio for the year 2010-11 is 18.02%

(iii) **Calculation of Standard Deviation**

M Ltd.

Exp. market value	*Exp. gain*	*Exp. div.*	*Exp Yield (1)*	*Prob. Factor (2)*	(1) X(2)	*Dev. (P_M- $\overline{P_M}$)*	*Square of dev. (3)*	(2) X (3)
220	0	20	20	0.2	4	-33	1089	217.80
250	30	20	50	0.5	25	-3	9	4.50
280	60	20	80	0.3	24	27	729	218.70
					53			σ^2_M = 441.00

Standard Deviation (σ_M) 21

N Ltd.

Exp. market value	*Exp. gain*	*Exp. div.*	*Exp Yield (1)*	*Prob. Factor (2)*	(1) X(2)	*Dev. (P_N- $\overline{P_N}$)*	*Square of dev. (3)*	(2) X (3)
290	0	3.5	3.5	0.2	0.7	-22	484	96.80
310	20	3.5	23.5	0.5	11.75	-2	4	2.00
330	40	3.5	43.5	0.3	13.05	18	324	97.20
					25.5			σ^2_N = 196.00

Standard Deviation (σ_N) 14

Alternatively based on return in percentage terms Standard Deviation can also be computed as follows:

M Ltd.

Exp. market value	*Exp. gain*	*Exp. div.*	*Exp Return (1)*	*Prob. Factor (2)*	(1) X(2)	*Dev. (P_M-$\overline{P_M}$)*	*Square of dev. (3)*	(2) X (3)
220	0	20	9.09	0.2	1.82	-15.01	225.30	45.06
250	30	20	22.73	0.5	11.37	-1.37	1.88	0.94
280	60	20	36.36	0.3	10.91	12.26	150.31	45.09
					24.10			σ^2_M = 91.09

Standard Deviation (σ_M) 9.54%

N Ltd.

Exp. market value	*Exp. gain*	*Exp. div.*	*Exp Return (1)*	*Prob. Factor (2)*	(1) X (2)	*Dev. (P_N-$\overline{P_N}$)*	*Square of dev. (3)*	(2) X (3)
290	0	3.5	1.21	0.2	0.24	-7.58	57.46	11.49
310	20	3.5	8.10	0.5	4.05	-0.69	0.48	0.24
330	40	3.5	15.00	0.3	4.50	6.21	38.56	11.57
					8.79			σ^2_N = 23.30

Standard Deviation (σ_N) 4.83%

Share of company M Ltd. is more risky as the S.D. is more than company N Ltd.

Practical Q33

33. An investor holds two stocks A and B. An analyst prepared ex-ante probability distribution for the possible economic scenarios and the conditional returns for two stocks and the market index as shown below:

Economic scenario	Probability	Conditional Returns %		
		A	B	Market
Growth	0.40	25	20	18
Stagnation	0.30	10	15	13
Recession	0.30	-5	-8	-3

The risk free rate during the next year is expected to be around 11%. Determine whether the investor should liquidate his holdings in stocks A and B or on the contrary make fresh investments in them. CAPM assumptions are holding true.

Solution — ICAI

33. Expected Return on stock A = E (A) = $\sum_{i=G,S,R} P_i A_i$

(G,S & R, denotes Growth, Stagnation and Recession)

(0.40)(25) + 0.30(10)+ 0.30(-5) = 11.5%

Expected Return on 'B'

(0.40×20) + (0.30×15) +0.30× (-8)=10.1%

Expected Return on Market index

(0.40 × 18) + (0.30 × 13) + 0.30 × (-3) =10.2%

Variance of Market index

$(18 - 10.2)^2 (0.40) + (13 - 10.2)^2 (0.30) + (-3 - 10.2)^2 (0.30)$

= 24.34 + 2.35 + 52.27 = 78.96%

Covariance of stock A and Market Index M

Cov. (AM) = $\sum_{i=G,S,R} ([A_i - E(A)][M_i - E(M)]P$

(25 -11.5) (18 - 10.2)(0.40) + (10 - 11.5) (13 - 10.2) (0.30) + (-5-11.5) (-3-10.2)(0.30)

= 42.12 + (-1.26) + 65.34=106.20

Covariance of stock B and Market index M

(20-10.1) (18-10.2)(0.40)+(15-10.1)(13-10.2)(0.30) + (-8-10.1)(-3-10.2)(0.30)= 30.89 + 4.12 + 71.67=106.68

Beta for stock A = $\frac{CoV(AM)}{VAR(M)} = \frac{106.20}{78.96} = 1.345$

Beta for Stock B = $\frac{CoV(BM)}{VarM} = \frac{106.68}{78.96} = 1.351$

Required Return for A

$R(A) = R_f + \beta (M - R_f)$

11% + 1.345(10.2 - 11) % = 9.924%

Required Return for B

11% + 1.351 (10.2 – 11) % = 9.92%

Alpha for Stock A

E (A) – R (A) i.e. 11.5 % – 9.924% = 1.576%

Alpha for Stock B

E (B) – R (B) i.e. 10.1% - 9.92% = 0.18%

Since stock A and B both have positive Alpha, therefore, they are underpriced. The investor should make fresh investment in them.

Practical Q34

34. Following are the details of a portfolio consisting of three shares:

Share	Portfolio weight	Beta	Expected return in %	Total variance
A	0.20	0.40	14	0.015
B	0.50	0.50	15	0.025
C	0.30	1.10	21	0.100

Standard Deviation of Market Portfolio Returns = 10%

You are given the following additional data:

Covariance (A, B) = 0.030

Covariance (A, C) = 0.020

Covariance (B, C) = 0.040

Calculate the following:

(i) The Portfolio Beta

(ii) Residual variance of each of the three shares

(iii) Portfolio variance using Sharpe Index Model

(iv) Portfolio variance (on the basis of modern portfolio theory given by Markowitz)

Solution — ICAI

34. **(i)** Portfolio Beta

0.20 x 0.40 + 0.50 x 0.50 + 0.30 x 1.10 = 0.66

(ii) Residual Variance

To determine Residual Variance first of all we shall compute the Systematic Risk as follows:

$\beta_A^2 \times \sigma_M^2 = (0.40)^2(0.01) = 0.0016$

$\beta_B^2 \times \sigma_M^2 = (0.50)^2(0.01) = 0.0025$

$\beta_C^2 \times \sigma_M^2 = (1.10)^2(0.01) = 0.0121$

Residual Variance

A 0.015 – 0.0016 = 0.0134

B 0.025 – 0.0025 = 0.0225

C 0.100 – 0.0121 = 0.0879

(iii) Portfolio variance using Sharpe Index Model

Systematic Variance of Portfolio = $(0.10)^2$ x $(0.66)^2$ = 0.004356

Unsystematic Variance of Portfolio = 0.0134 x $(0.20)^2$ + 0.0225 x $(0.50)^2$ + 0.0879 x $(0.30)^2$ = 0.014072

Total Variance = 0.004356 + 0.014072 = 0.018428

(iv) Portfolio variance on the basis of Markowitz Theory

$= (w_A \times w_A \times \sigma_A^2) + (w_A \times w_B \times Cov_{AB}) + (w_A \times w_C \times Cov_{AC}) + (w_B \times w_A \times Cov_{AB}) + (w_B \times w_B \times \sigma_B^2) + (w_B \times w_C \times Cov_{BC}) + (w_C \times w_A \times Cov_{CA}) + (w_C \times w_B \times Cov_{CB}) + (w_C \times w_C \times \sigma_C^2)$

= (0.20 x 0.20 x 0.015) + (0.20 x 0.50 x 0.030) + (0.20 x 0.30 x 0.020) + (0.20 x 0.50 x 0.030) + (0.50 x 0.50 x 0.025) + (0.50 x 0.30 x 0.040) + (0.30 x 0.20 x 0.020) + (0.30 x 0.50 x 0.040) + (0.30 x 0.30 x 0.10)

= 0.0006 + 0.0030 + 0.0012 + 0.0030 + 0.00625 + 0.0060 + 0.0012 + 0.0060 + 0.0090

= 0.0363

Practical Q35

35. Ramesh wants to invest in stock market. He has got the following information about individual securities:

Security	Expected Return	Beta	σ^2_{ei}
A	15	1.5	40
B	12	2	20
C	10	2.5	30
D	09	1	10
E	08	1.2	20
F	14	1.5	30

Market index variance is 10 percent and the risk free rate of return is 7%. What should be the optimum portfolio assuming no short sales?

Solution — ICAI

35. Securities need to be ranked on the basis of excess return to beta ratio from highest to the lowest.

Security	R_i	β_i	$R_i - R_f$	$\frac{R_i - R_f}{\beta_i}$
A	15	1.5	8	5.33
B	12	2	5	2.5
C	10	2.5	3	1.2
D	9	1	2	2
E	8	1.2	1	0.83
F	14	1.5	7	4.67

Ranked Table:

Security	$R_i - R_f$	β_i	σ^2_{ei}	$\frac{(R_i - R_f) \times \beta_i}{\sigma^2_{ei}}$	$\sum_{e=i}^{N} \frac{(R_i - R_f) \times \beta_i}{\sigma^2_{ei}}$	$\frac{\beta_i^2}{\sigma^2_{ei}}$	$\sum_{e=i}^{N} \frac{\beta_i^2}{\sigma^2_{ei}}$	C_i
A	8	1.5	40	0.30	0.30	0.056	0.056	1.923
F	7	1.5	30	0.35	0.65	0.075	0.131	2.814
B	5	2	20	0.50	1.15	0.20	0.331	2.668
D	2	1	10	0.20	1.35	0.10	0.431	2.542
C	3	2.5	30	0.25	1.60	0.208	0.639	2.165
E	1	1.2	20	0.06	1.66	0.072	0.711	2.047

CA = 10 x 0.30 / [1 + (10 x 0.056)] = 1.923

CF = 10 x 0.65 / [1 + (10 x 0.131)] = 2.814

CB = 10 x 1.15 / [1 + (10 x 0.331)] = 2.668

CD = 10 x 1.35 / [1 + (10 x 0.431)] = 2.542

CC = 10 x 1.60 / [1 + (10 x 0.639)] = 2.165

CE = 10 x 1.66 / [1 + (10 x 0.7111)] = 2.047

Cut off point is 2.814

$$Z_i = \frac{\beta_i}{\sigma^2_{ei}}\left[\left(\left[\left[\frac{(R_i - R_f)}{\beta_i} - C\right]\right]\right)\right]$$

$$Z_A = \frac{1.5}{40}(5.33 - 2.814) = 0.09435$$

$$Z_F = \frac{1.5}{30}(4.67 - 2.814) = 0.0928$$

$$X_A = 0.09435/[0.09435 + 0.0928] = 50.41\%$$

$$X_F = 0.0928/[0.09435 + 0.0928] = 49.59\%$$

Funds to be invested in security A & F are 50.41% and 49.59% respectively.

Practical Q36

36. A Portfolio Manager (PM) has the following four stocks in his portfolio:

Security	No. of Shares	Market Price per share (₹)	β
VSL	10,000	50	0.9
CSL	5,000	20	1.0
SML	8,000	25	1.5
APL	2,000	200	1.2

Compute the following:

(i) Portfolio beta.

(ii) If the PM seeks to reduce the beta to 0.8, how much risk free investment should he bring in?

(iii) If the PM seeks to increase the beta to 1.2, how much risk free investment should he bring in?

Solution — ICAI

36. (i)

Security	No. of shares (1)	Market Price of Per Share (2)	(1) × (2)	% to total (w)	ß (x)	wx
VSL	10000	50	500000	0.4167	0.9	0.375
CSL	5000	20	100000	0.0833	1	0.083
SML	8000	25	200000	0.1667	1.5	0.250
APL	2000	200	400000	0.3333	1.2	0.400
			1200000	1		1.108

Portfolio beta 1.108

(ii) Required Beta 0.8

It should become (0.8 / 1.108) 72.2 % of present portfolio

If ₹ 12,00,000 is 72.20%, the total portfolio should be

₹ 12,00,000 × 100/72.20 or ₹ 16,62,050

Additional investment in zero risk should be (₹ 16,62,050 – ₹ 12,00,000) = ₹ 4,62,050

Revised Portfolio will be

Security	No. of shares (1)	Market Price of Per Share (2)	(1) × (2)	% to total (w)	ß (x)	wx
VSL	10000	50	500000	0.3008	0.9	0.271
CSL	5000	20	100000	0.0602	1	0.060
SML	8000	25	200000	0.1203	1.5	0.180
APL	2000	200	400000	0.2407	1.2	0.289

Risk free asset	46205	10	462050	0.2780	0	0
			1662050	1		0.800

(iii) To increase Beta to 1.2

Required beta 1.2

It should become 1.2 / 1.108 108.30% of present beta

If 1200000 is 108.30%, the total portfolio should be

1200000 × 100/108.30 or 1108033 say 1108030

Additional investment should be (-) 91967 i.e. Divest ₹ 91970 of Risk Free Asset

Revised Portfolio will be

Security	No. of shares (1)	Market Price of Per Share (2)	(1) × (2)	% to total (w)	ß (x)	wx
VSL	10000	50	500000	0.4513	0.9	0.406
CSL	5000	20	100000	0.0903	1	0.090
SML	8000	25	200000	0.1805	1.5	0.271
APL	2000	200	400000	0.3610	1.2	0.433
Risk free asset	-9197	10	-91970	-0.0830	0	0
			1108030	1		1.20

Portfolio beta 1.20

Alternative Approach

(ii) Let x be the amount of Risk-Free Asset to be acquired, then

Security	(1) × (2)	ß (x)	wx
VSL	500000	0.9	450000
CSL	100000	1	100000
SML	200000	1.5	300000
APL	400000	1.2	480000
Risk free asset	x	0	0
	1200000 + x		1330000

Accordingly,

$$\frac{13,30,000}{12,00,000+x} = 0.8$$

x = 462500 i.e. value of Risk Free Asset to be purchased to decrease beta of portfolio to 0.8.

(iii) Similarly let y the amount of Risk Free Assets to be divest, then

$$\frac{13,30,000}{12,00,000+y} = 1.20$$

y = -91,667 i.e. value of Risk Free Asset to be divested to increase beta of portfolio to 1.20.

Practical Q37

37. A has portfolio having following features:

Security	β	Random Error σ_{ei}	Weight
L	1.60	7	0.25
M	1.15	11	0.30
N	1.40	3	0.25
K	1.00	9	0.20

You are required to find out the risk of the portfolio if the standard deviation of the market index (σ_m) is 18%.

Solution — ICAI

37. $\beta_p = \sum_{i=1} x_i\beta_i$

$= 1.60 \times 0.25 + 1.15 \times 0.30 + 1.40 \times 0.25 + 1.00 \times 0.20$

$= 0.4 + 0.345 + 0.35 + 0.20 = 1.295$

The Standard Deviation (Risk) of the portfolio is

$= [(1.295)^2(18)^2+(0.25)^2(7)^2+(0.30)^2(11)^2+(0.25)^2(3)^2+(0.20)^2(9)^2)]^{½}$

$= [543.36 + 3.0625 + 10.89 + 0.5625 + 3.24]^{½} = [561.115]^{½} = 23.69\%$

Alternative Answer

The variance of Security's Return

$\sigma^2 = \beta_i^2 \sigma^2_m + \sigma^2_{ei}$

Accordingly, variance of various securities

		σ^2	Weight(w)	σ^2Xw
L	$(1.60)^2 (18)^2 + 7^2$ =	878.44	0.25	219.61
M	$(1.15)^2 (18)^2 + 11^2$ =	549.49	0.30	164.85
N	$(1.40)^2 (18)^2 + 3^2$ =	644.04	0.25	161.01
K	$(1.00)^2 (18)^2 + 9^2$ =	405.00	0.20	81
		Variance		626.47

SD = $\sqrt{626.47}$ = 25.03

Practical Q38

38. Mr. Tamarind intends to invest in equity shares of a company the value of which depends upon various parameters as mentioned below:

Factor	Beta	Expected value in %	Actual value in %
GNP	1.20	7.70	7.70
Inflation	1.75	5.50	7.00
Interest rate	1.30	7.75	9.00
Stock market index	1.70	10.00	12.00
Industrial production	1.00	7.00	7.50

If the risk free rate of interest be 9.25%, how much is the return of the share under Arbitrage Pricing Theory?

Solution — ICAI

38. Return of the stock under APT

Factor	Actual value in %	Expected value in %	Difference	Beta	Diff. x Beta
GNP	7.70	7.70	0.00	1.20	0.00
Inflation	7.00	5.50	1.50	1.75	2.63
Interest rate	9.00	7.75	1.25	1.30	1.63
Stock index	12.00	10.00	2.00	1.70	3.40
Ind. Production	7.50	7.00	0.50	1.00	0.50
					8.16
Risk free rate in %					9.25
Return under APT					17.41

Practical Q39

39. The total market value of the equity share of O.R.E. Company is ₹ 60,00,000 and the total value of the debt is ₹ 40,00,000. The treasurer estimate that the beta of the stock is currently 1.5 and that the expected risk premium on the market is 10 per cent. The treasury bill rate is 8 per cent.

Required:

(i) What is the beta of the Company's existing portfolio of assets?

(ii) Estimate the Company's Cost of capital and the discount rate for an expansion of the company's present business.

Solution — ICAI

39. (i) $$\beta_{asset} = \beta_{equity} \times \frac{V_E}{V_0} + B_{debt} \times \frac{V_D}{V_0}$$

Note: Since β_{debt} is not given it is assumed that company debt capital is virtually riskless.

If company's debt capital is riskless than above relationship become:

Here $\beta_{equity} = 1.5$; $\beta_{asset} = \beta_{equity} \frac{V_E}{V_0}$

As $\beta_{debt} = 0$

V_E = ₹ 60 lakhs.

V_D = ₹ 40 lakhs.

V_0 = ₹ 100 lakhs.

$$\beta_{asset} = 1.5 \times \frac{60 \text{ lakhs}}{100 \text{ lakhs}}$$

$$= 0.9$$

(ii) (a) If only equity is used to finance the expansion, the Cost of Capital for discounting company's expansion of existing business shall be computed as follows:

Company's cost of equity = $R_f + \beta_A \times$ Market Risk premium

Where R_f = Risk free rate of return

β_A = Beta of company assets

Therefore, company's cost of equity = 8% + 0.9 × 10 = 17% and overall cost of capital shall be 17%.

(b) Alternatively, if funds expansion are raised for in same proportion as exiting capital structure, then cost of capital shall be computed as follows:

Cost of Equity = 8% + 1.5 x 10 = 23%

Cost of Debt = 8%

WACC (Cost of Capital) = $23\% \times \frac{3}{5} + 8\% \times \frac{2}{5} = 17\%$

Practical Q40

40. Mr. Nirmal Kumar has categorized all the available stock in the market into the following types:

(i) Small cap growth stocks

(ii) Small cap value stocks

(iii) Large cap growth stocks

(iv) Large cap value stocks

Mr. Nirmal Kumar also estimated the weights of the above categories of stocks in the market index. Further, the sensitivity of returns on these categories of stocks to the three important factor are estimated to be:

Category of Stocks	Weight in the Market Index	Factor I (Beta)	Factor II (Book Price)	Factor III (Inflation)
Small cap growth	25%	0.80	1.39	1.35
Small cap value	10%	0.90	0.75	1.25
Large cap growth	50%	1.165	2.75	8.65
Large cap value	15%	0.85	2.05	6.75
Risk Premium		6.85%	-3.5%	0.65%

The rate of return on treasury bonds is 4.5%

Required:

(a) Using Arbitrage Pricing Theory, determine the expected return on the market index.

(b) Using Capital Asset Pricing Model (CAPM), determine the expected return on the market index.

(c) Mr. Nirmal Kumar wants to construct a portfolio constituting only the 'small cap value' and 'large cap growth' stocks. If the target beta for the desired portfolio is 1, determine the composition of his portfolio.

Solution — ICAI

40. (a) Method I

Stock's return

Small cap growth = 4.5 + 0.80 x 6.85 + 1.39 x (-3.5) + 1.35 x 0.65 = 5.9925%

Small cap value = 4.5 + 0.90 x 6.85 + 0.75 x (-3.5) + 1.25 x 0.65 = 8.8525%

Large cap growth = 4.5 + 1.165 x 6.85 + 2.75 x (-3.5) + 8.65 x 0.65 = 8.478%

Large cap value = 4.5 + 0.85 x 6.85 + 2.05 x (-3.5) + 6.75 x 0.65 = 7.535%

Expected return on market index

0.25 x 5.9925 + 0.10 x 8.8525 + 0.50 x 8.478 + 0.15 x 7.535 = 7.7526%

Method II

Expected return on the market index

= 4.5% + [0.1x0.9 + 0.25x0.8 + 0.15x0.85 + 0.50x1.165] x 6.85 + [(0.75 x 0.10 + 1.39 x 0.25 + 2.05 x 0.15 + 2.75 x 0.5)] x (-3.5) + [{1.25 x 0.10 + 1.35 x 0.25 + 6.75 x 0.15 + 8.65 x 0.50)] x 0.65

= 4.5 + 6.85 + (-7.3675) + 3.77 = 7.7525%.

(b) Using CAPM,

Small cap growth = 4.5 + 6.85 x 0.80 = 9.98%

Small cap value = 4.5 + 6.85 x 0.90 = 10.665%

Large cap growth = 4.5 + 6.85 x 1.165 = 12.48%

Large cap value = 4.5 + 6.85 x 0.85 = 10.3225%

Expected return on market index

= 0.25 x 9.98 + 0.10 x 10.665 + 0.50 x 12.45 + 0.15 x 10.3225 = 11.33%

(c) Let us assume that Mr. Nirmal will invest X_1% in small cap value stock and X_2% in large cap growth stock

$X_1 + X_2 = 1$

$0.90\ X_1 + 1.165\ X_2 = 1$

$0.90\ X_1 + 1.165(1 - X_1) = 1$

$0.90\ X_1 + 1.165 - 1.165\ X_1 = 1$

$0.165 = 0.265\ X_1$

$$\frac{0.165}{0.265} = X_1$$

$0.623 = X_1$, $X_2 = 0.377$

62.3% in small cap value

37.7% in large cap growth.

Practical Q41

41. The following are the data on five mutual funds:

Fund	Return	Standard Deviation	Beta
A	15	7	1.25
B	18	10	0.75
C	14	5	1.40
D	12	6	0.98
E	16	9	1.50

You are required to compute Reward to Volatility Ratio and rank these portfolio using:

- Sharpe method and
- Treynor's method

assuming the risk free rate is 6%.

Solution — ICAI

41. Sharpe Ratio $S = (R_p - R_f)/\sigma_p$

Treynor Ratio $T = (R_p - R_f)/\beta_p$

Where,

R_p = Return on Fund

R_f = Risk-free rate

σ_p = Standard deviation of Fund

β_p = Beta of Fund

Reward to Variability (Sharpe Ratio)

Mutual Fund	R_p	R_f	$R_p - R_f$	σ_p	Reward to Variability	Ranking
A	15	6	9	7	1.285	2
B	18	6	12	10	1.20	3
C	14	6	8	5	1.60	1
D	12	6	6	6	1.00	5
E	16	6	10	9	1.11	4

Reward to Volatility (Treynor Ratio)

Mutual Fund	R_p	R_f	$R_p - R_f$	β_p	Reward to Volatility	Ranking
A	15	6	9	1.25	7.2	2
B	18	6	12	0.75	16	1
C	14	6	8	1.40	5.71	5
D	12	6	6	0.98	6.12	4
E	16	6	10	1.50	6.67	3

Practical Q42

42. Five portfolios experienced the following results during a 7- year period:

Portfolio	Average Annual Return (R_p) (%)	Standard Deviation (S_p)	Correlation with the market returns (r)
A	19.0	2.5	0.840
B	15.0	2.0	0.540
C	15.0	0.8	0.975
D	17.5	2.0	0.750
E	17.1	1.8	0.600
Market Risk (σ_m)		1.2	
Market rate of Return (R_m)	14.0		
Risk-free Rate (R_f)	9.0		

Rank the portfolios using (a) Sharpe's method, (b) Treynor's method and (c) Jensen's Alpha

Solution — ICAI

42. Let portfolio standard deviation be σ_p

Market Standard Deviation = σ_m

Coefficient of correlation = r

Portfolio beta (β_p) = $\frac{\sigma_p r}{\sigma_m}$

Required portfolio return (R_p) = $R_f + \beta_p (R_m - R_f)$

Portfolio	Beta	Return from the portfolio (R_p) (%)
A	1.75	17.75
B	0.90	13.50
C	0.65	12.25
D	1.25	15.25
E	0.90	13.50

Portfolio	Sharpe Method		Treynor Method		Jensen's Alpha	
	Ratio	Rank	Ratio	Rank	Ratio	Rank
A	4.00	IV	5.71	V	1.25	V
B	3.00	V	6.67	IV	1.50	IV
C	7.50	I	9.23	I	2.75	II
D	4.25	III	6.80	III	2.25	III
E	4.50	II	9.00	II	3.60	I

TYK practical with ICAI solutions

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